IE437 Data-Driven Decision Making and Control
2023 Spring Semester

2. Fundamentals on Bayesian Statistics

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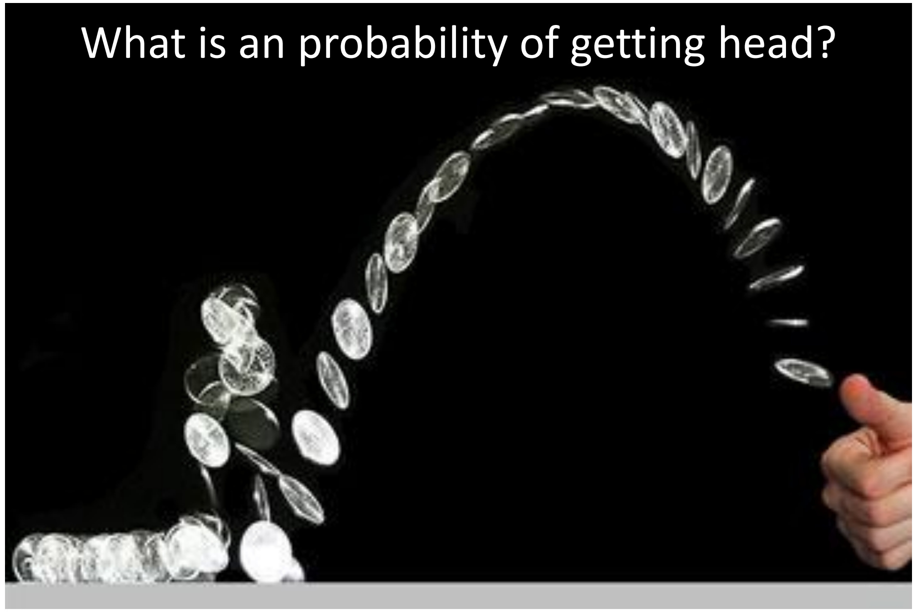
Fundamentals on Bayesian Statistics

Introduction



Statistics infer the causes that generated the observed data.

Model → data

θ (parameters) :
characteristics of a model

$y = (y_1, \ldots, y_n)$:
Observed consequence

θ : Probability of having a head for each coin tossing

$(\text{Head}, \text{Head}, \text{Tail}, \ldots)$

Introduction

- **View on Probability**

- **Frequentists**:
  - Probability only has meaning in terms of a limiting case of repeated measurements
  - Probabilities are fundamentally related to frequencies of events.

- **Bayesian**:
  - Degrees of certainty about statements
  - Probabilities are fundamentally related to our own knowledge about an event.

Introduction

- **View on Statistics**

- **Frequentists**:
  - **Data are a repeatable random sample** → there is a frequency of occurrence
  - Underlying parameters remain constant during this repeatable process
  - **Parameters are fixed and unchanging** under all realistic circumstances

- **Bayesian**:
  - Data are observed from the realized sample
  - **Parameters are unknown and described probabilistically**
    (View the world probabilistically)
  - **Data are fixed**

Introduction

- **Frequentist vs Bayesian: Coin Tossing**

- **Frequentist**
  - θ is relative frequency of head in a "large number" of "identical flip"
  - Statistical results assume that data were from a controlled experiment
  - Nothing is more important than repeatability

Try

1. Estimate the parameter θ by conducting experiments
2. Give me estimates

- **Issues**
  - When the number of trials n is small, estimation is biased

$$\frac{\#\text{Success}}{\#\text{Trials}} = \frac{1}{3}, \quad \frac{5}{6}, \quad \frac{5}{13}, \quad \frac{129}{313}, \quad \frac{61423}{123400}$$

  - Identical flip (controlled experiment) is unrealistic

Introduction

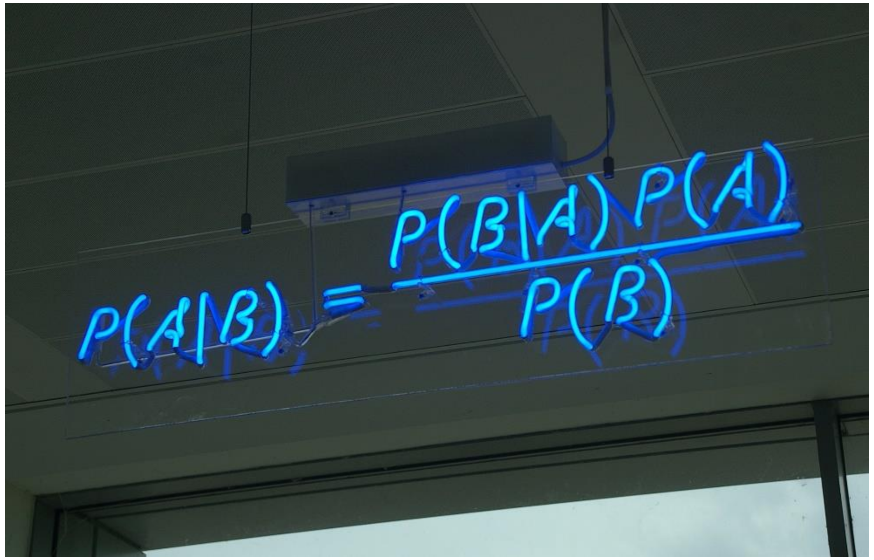
- **Frequentist vs Bayesian: Coin Tossing**

- **Frequentist**
  - θ is relative frequency of head in a "large number" of "identical flip"
  - Statistical results assume that data were from a controlled experiment
  - Nothing is more important than repeatability

- **Bayesian**
  - Parameters θ are varied (uncertain) $\leftarrow$ Core part of Bayesian approach
  - Use probability concept to provide our belief on θ: $p(\theta)$
  - Each parameter θ can be associated with different conditions, i.e., orientation, force, etc.
  - Data fixed

- **Issues**
  - Subjective on $p(\theta)$
  - How to specify $p(\theta)$

Bayes' Rule



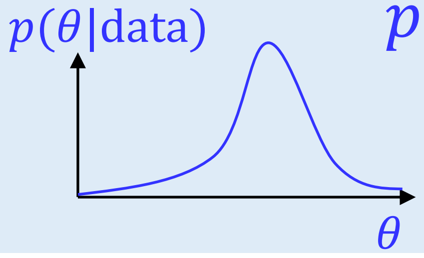
- Bayes' rule will play fundamental roles in formulating various machine learning problems

- Derivation of Bayes' rule:

$$\boxed{\begin{gathered} P(A|B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B|A) = \frac{P(B \cap A)}{P(A)} \\ P(A \cap B) = P(B \cap A) \Rightarrow P(A|B)P(B) = P(B|A)P(A) \\ \Rightarrow P(A|B) = \frac{P(B|A)P(A)}{P(B)} \end{gathered}}$$

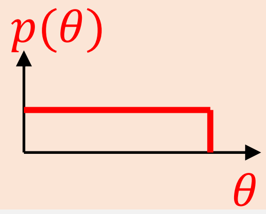
Bayes' rule in Statistics

The posterior :
The probability on the parameter θ given the evidence (data)



The evidence
The probability of getting this evidence given the parameter

The Prior
The belief (uncertainty) about the parameter θ



$$p(\theta|\text{data}) = \frac{p(\text{data}|\theta)p(\theta)}{p(\text{data})}$$

The Marginal probability
The probability of the evidence, Probability of data over all possibilities

- Parameter instantiates one model among a model class

$$p(\theta|\text{data}, \text{model}) = \frac{p(\text{data}|\theta, \text{model})p(\theta|\text{model})}{p(\text{data}|\text{model})}$$

Approaches

1. Select a model for data (structure)
2. Specify a prior on model parameters
3. Specify Likelihood
4. Construct posterior
5. If necessary, predict unobserved value given the updated information on the parameters

Estimating Model Parameters

Frequentist approach

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$
$$= p(y|\theta)k(y)$$
$$\propto p(y|\theta)$$

$$\left(k(y) = \frac{p(\theta)}{p(y)}\right)$$

$$L(\theta) = p(y|\theta)$$

- θ is fixed
- Maximize **likelihood** to estimate θ (single point estimation)

$$\theta^* = \underset{\theta}{\operatorname{argmax}}\, L(\theta)$$

Maximum Likelihood estimation (MLE)

Bayesian approach

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$
$$= \frac{p(y|\theta)p(\theta)}{\int_\theta p(y|\theta)p(\theta)d\theta}$$
$$\propto p(y|\theta)p(\theta)$$

- θ is random
- Estimation is expressed as prob. dist.
- Posterior is a ***balance*** between our belief and observation

In case we need to pick up one parameter:

$$\theta^* = \underset{\theta}{\operatorname{argmax}}\, p(\theta|y)$$

Maximum a posteriori estimation (MAP)

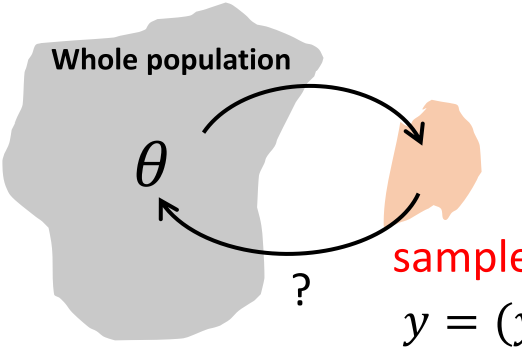
Maximum Likelihood Estimation for Coin Flipping Probability

- Coin tossing models:

$$Y_i = \begin{cases} 1 & \text{if Head} \\ 0 & \text{if Tail} \end{cases}$$

$$p(y_i|\theta) = \mathrm{B}(y_i|\theta) = \theta^{y_i}(1-\theta)^{1-y_i}$$

$\theta \in [0,1]$: Probability of having a head



- Likelihood of a single observation:

$$L(\theta) = p(y_i|\theta) = \theta^{y_i}(1-\theta)^{1-y_i}$$

- Likelihood of a three-observations $y = (y_1, y_2, y_3) = (1, 0, 1)$:

$$L(\theta) = p(y_1 = 1, y_2 = 0, y_3 = 1|\theta) = p(1|\theta)p(0|\theta)p(1|\theta) = \theta^2(1-\theta)^1$$

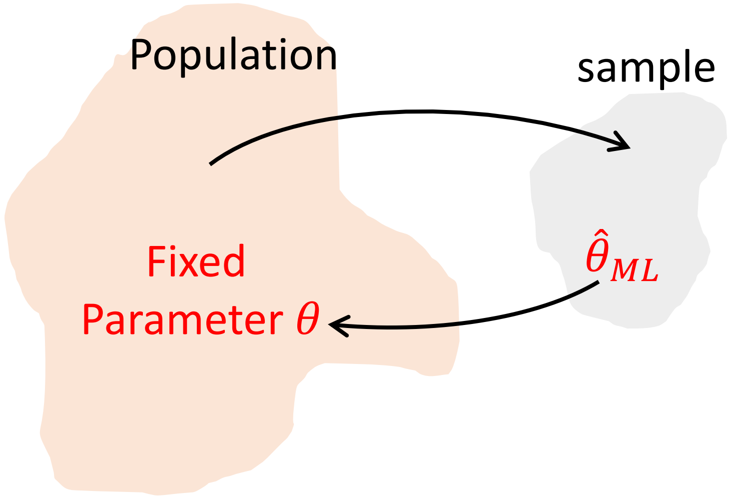
i.i.d.→ Exchangeability

- Likelihood of n-observations:

$$L(\theta) = p(y_1, y_2, \cdots, y_n|\theta) = \theta^{\sum y_i}(1-\theta)^{n-\sum y_i}$$

$\sum y_i$: Number of Heads in n trials (sufficient statistics)

Maximum Likelihood Estimation for Coin Flipping Probability



- Maximum likelihood estimation:

$$\hat{\theta}_{\text{ML}} = \underset{\theta}{\operatorname{argmax}}\, L(\theta) = p(y_1, y_2, \cdots, y_n | \theta) = \theta^{\sum y_i}(1-\theta)^{n-\sum y_i}$$

$$\frac{dL(\theta)}{d\theta} = \sum y_i\, \theta^{\sum y_i - 1}(1-\theta)^{n-\sum y_i} - \theta^{\sum y_i}\left(n - \sum y_i\right)(1-\theta)^{n-\sum y_i - 1} = 0$$

$$\Rightarrow \hat{\theta}_{\text{ML}} = \frac{\sum y_i}{n}$$

MLE estimation gives a **relative frequency**

Bayesian Approach for Estimating Model Parameters

- The essential characteristics of Bayesian methods
  = explicit use of probability for **quantifying uncertainty** in the statistical model

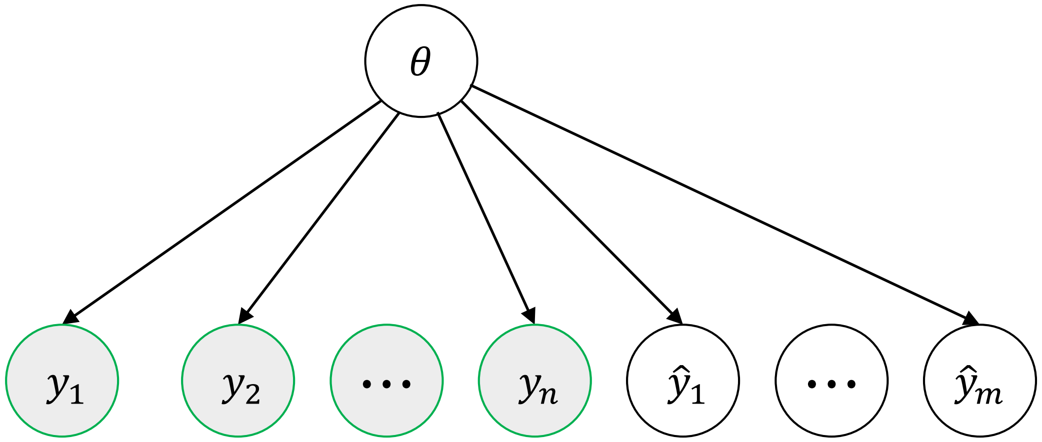
- Bayes rule:

$$
\begin{aligned}
p(\theta|y) &= \frac{p(y|\theta)p(\theta)}{p(y)} \\
&= \frac{p(y|\theta)p(\theta)}{\int_\theta p(y|\theta)p(\theta)d\theta} \qquad \left(\because p(y) = \int_\theta p(y|\theta)p(\theta)d\theta\right) \\
&\propto p(y|\theta)p(\theta)
\end{aligned}
$$

Posterior ∝ Likelihood × Prior

- $p(\theta)$: Prior – subjective belief about θ
- $p(y|\theta)$: Likelihood – observation (data) regarding θ
- $p(\theta|y)$: Posterior – updated belief about θ with the data

Bayesian Inference Problems



$y = (y_1, \dots, y_n)$
Observed

$\hat{y} = (\hat{y}_1, \dots, \hat{y}_m)$
Unobserved (future value)

- Prior predictive distribution

$$p(y) = \int_\theta p(y, \theta) d\theta = \int_\theta p(y|\theta) p(\theta) d\theta$$

- Posterior distribution

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{\int_\theta p(y|\theta)p(\theta)d\theta} \propto p(y|\theta)p(\theta)$$

- Posterior predictive distribution

$$p(\hat{y}|y) = \int_\theta p(\hat{y}, \theta|y) d\theta = \int_\theta p(\hat{y}|\theta, y) p(\theta|y) d\theta = \int_\theta p(\hat{y}|\theta) p(\theta|y) d\theta$$

$p(\hat{y}|\theta, y) = p(\hat{y}|\theta)$ because $\hat{y} \perp y |\ \theta$

Bayesian Estimation for Coin Flipping Probability

- Coin tossing models:

Bernoulli distribution

Binomial distribution

n times

$$Y_i = \begin{cases} 1 & \text{if Head} \\ 0 & \text{if Tail} \end{cases}$$

$$Y_i \sim \mathrm{B}(\theta)$$

$$P(y_i|\theta) = \mathrm{B}(y_i|\theta) = \theta^{y_i}(1-\theta)^{1-y_i}$$

$\theta \in [0,1]$: Probability of having a head

$Y \in \{0,1,\ldots,n\}$: The number heads

$$Y \sim \mathrm{Bin}(n,\theta)$$

$$p(y|\theta) = \mathrm{Bin}(y|n,\theta) = \binom{n}{y}\theta^{y}(1-\theta)^{n-y}$$

$\theta \in [0,1]$: Probability of having a head

Bayesian Estimation for Coin Flipping Probability

Likelihood :

$Y \sim \mathrm{Bin}(n, \theta) \rightarrow p(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$ Y is the number of success among n Bernoulli trial

Bayesian Estimation for Coin Flipping Probability

Likelihood :

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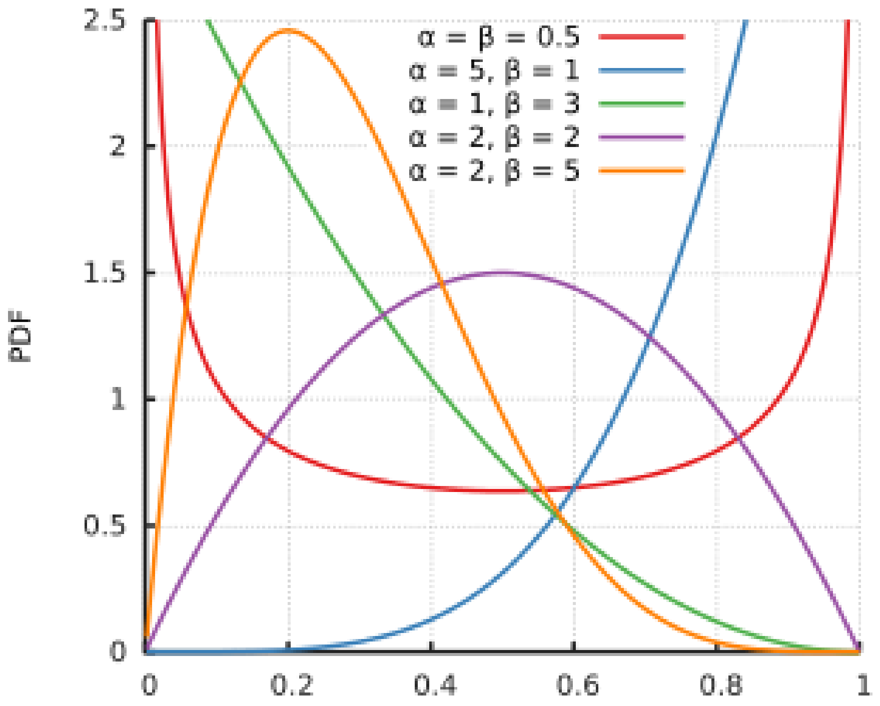
Prior:

$\theta \sim \mathrm{Beta}(\alpha, \beta) \rightarrow p(\theta) = \dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}$



Bayesian Estimation for Coin Flipping Probability

Likelihood :

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Prior:

$$\theta \sim \text{Beta}(\alpha, \beta) \rightarrow p(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

Posterior :

$$P(\theta|y) \propto P(y|\theta) p(\theta)$$

$$= \binom{n}{y} \theta^{y} (1-\theta)^{n-y} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$\propto \theta^{y} (1-\theta)^{n-y} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

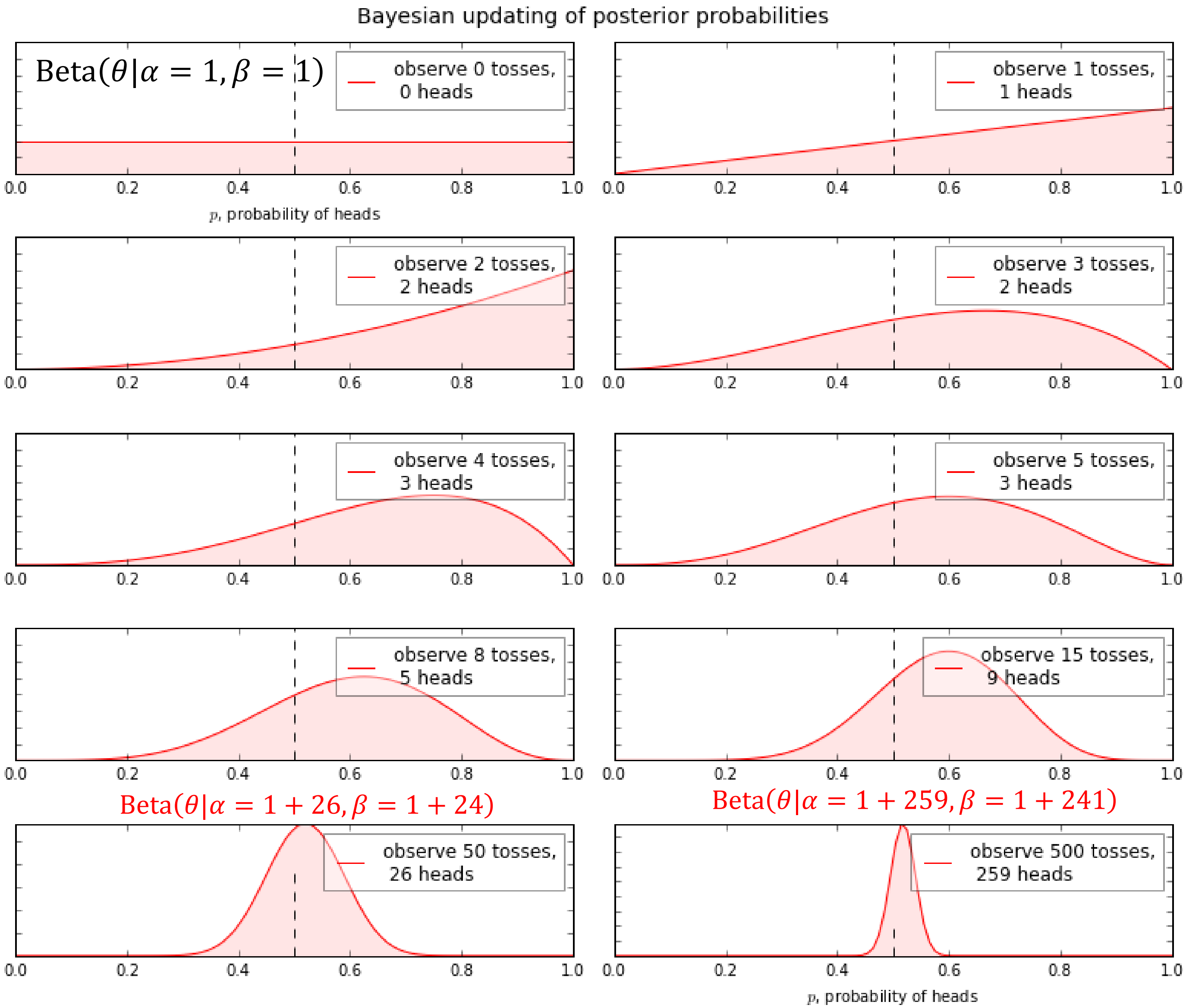
$$= \theta^{\alpha+y-1} (1-\theta)^{\beta+n-y-1}$$

$$= \text{Beta}(\theta | \alpha + y, \beta + n - y)$$

(α is pseudo counts for the success while β is a pseudo counts for the failure)

Bayesian Estimation for Coin Flipping Probability

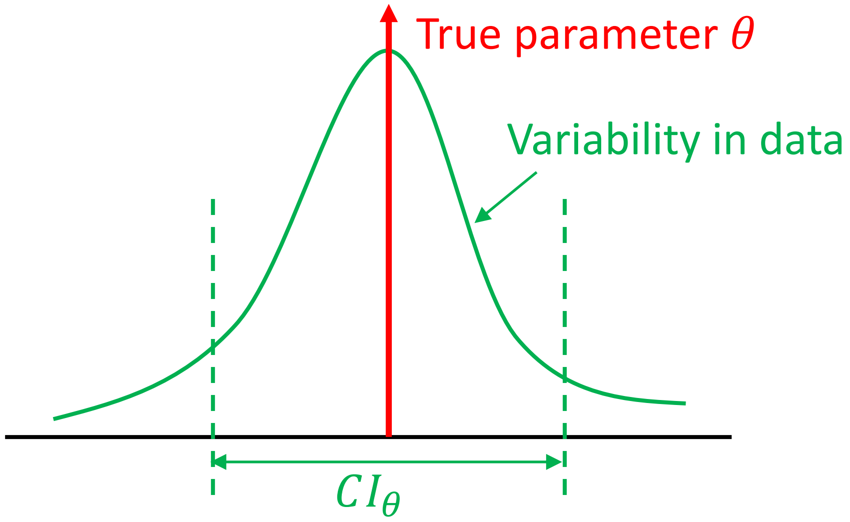
$\mathrm{Beta}(\theta|\alpha,\beta)$



Confidence Interval vs Credible region

Frequentist approach

- Describe variability in data given the fixed parameter θ

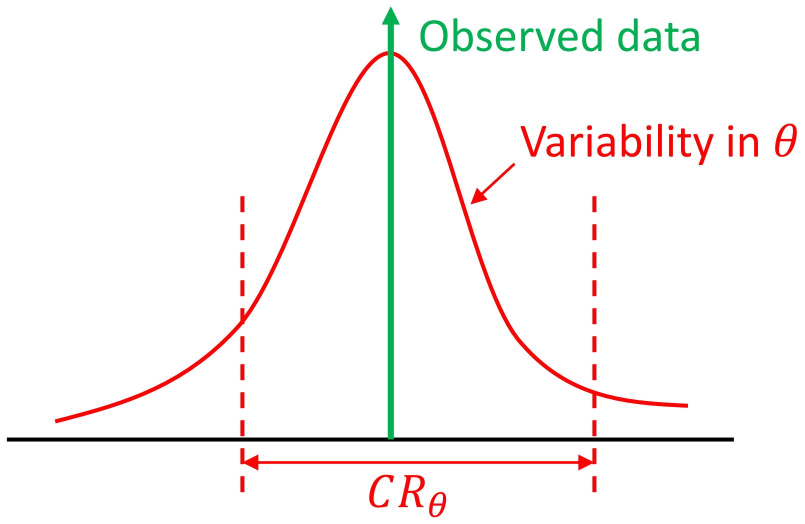


"There is a 95% probability that when I compute CI_θ from a current data, the computed CI contains θ_{true}

→ From current data set,
We can only say that $\theta \in \mathrm{CI}$ or $\theta \notin \mathrm{CI}$

Bayesian approach

- Describe variability in θ for fixed data



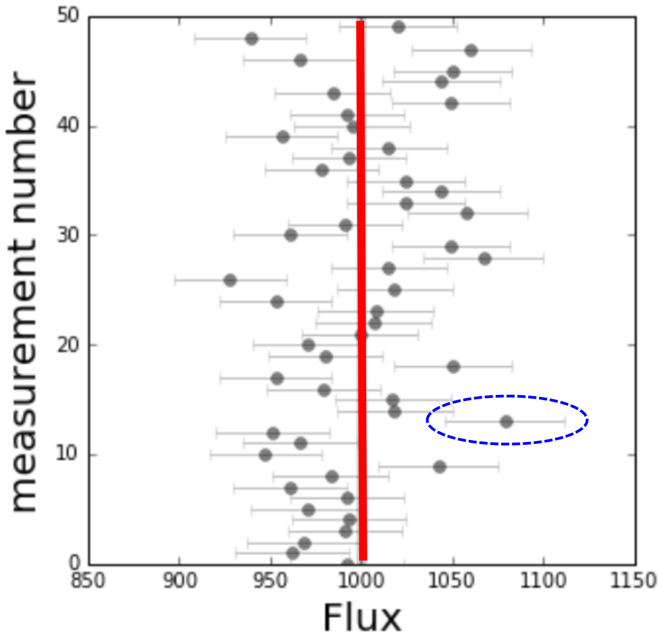
"Given our observed data, there is a 95% probability that the true value of θ falls within Credible region CR_θ"

→ From current data set,
We can make probabilistic statement such as $\Pr(\theta \in CR) = 0.95$

Confidence Interval vs Credible region

Frequentist approach

- Describe variability in data given the fixed parameter θ

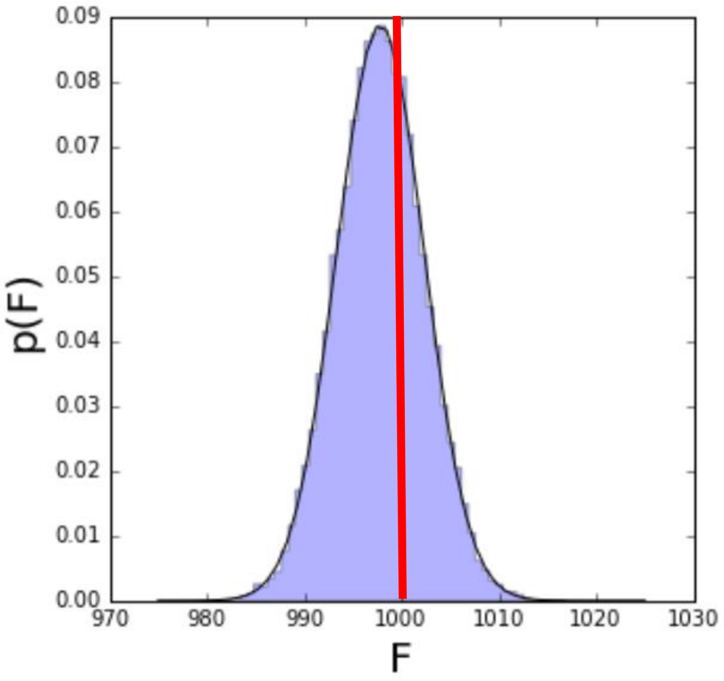


$\theta \in \mathrm{CI}$ or $\theta \notin \mathrm{CI}$

If the experiment were repeated an infinite number of times, 95% of the calculated intervals would contain θ .

Bayesian approach

- Describe variability in θ for fixed data



$\Pr(\theta \in \mathrm{CR}) = 0.95$

There is a 95% chance θ is in CR

Posterior as compromise between data and prior information

- From the previous result $p(\theta|y) = \text{Beta}(\theta|\alpha + y, \beta + n - y)$,

$$\mathbb{E}[\theta] = \frac{\alpha}{\alpha + \beta}$$

$$\mathbb{E}[\theta|y] = \frac{\alpha + y}{\alpha + \beta + n} = \frac{\alpha}{\alpha + \beta} \frac{\alpha + \beta}{(\alpha + \beta + n)} + \frac{y}{n} \frac{n}{(\alpha + \beta + n)}$$

$$= \mathbb{E}[\theta] \frac{\alpha + \beta}{\alpha + \beta + n} + \hat{\theta}_{ML} \frac{n}{\alpha + \beta + n}$$

- As $n \to \infty$, $\mathbb{E}[\theta|Y] \to \frac{y}{n} = \hat{\theta}_{\text{ML}}$

- Large value of $\alpha + \beta$ signifies Posterior

  ➢ In the limit, the prior does not influence the results. That is, the results are dominated by the data (observation).

$$\text{Var}(\theta|y) = \frac{(\alpha + y)(\beta + n - y)}{(\alpha + \beta + n)^2(\alpha + \beta + n + 1)} = \frac{\mathbb{E}[\theta|y](1 - \mathbb{E}[\theta|y])}{\alpha + \beta + n + 1}$$

- As n and $(n - y) \to \infty$, $\text{Var}(\theta|y) \to \frac{1}{n}\frac{y}{n}\left(1 - \frac{y}{n}\right) = \frac{p(1-p)}{n}$

Posterior as compromise between data and prior information

$$\mathbb{E}[\theta] = \mathbb{E}\big[\mathbb{E}[\theta|v]\big]$$

- **The prior mean of $\boldsymbol{\theta}$** is the average of all possible posterior means over the distribution of possible data

$$\mathrm{Var}(\theta) = \mathbb{E}[\mathrm{Var}(\theta|y)] + \mathrm{Var}(\mathbb{E}[\theta|y])$$

- **The posterior variance** is on average smaller than **the prior variance** by an amount that depends on the variation in **posterior means** over the distribution of all the possible data.

The posterior distribution is centered at a point that represents a compromise between the prior information and the data, and the compromise is controlled to a greater extent by the data as the sample size increases

Bayesian Estimation for Coin Flipping Probability

Likelihood :

$Y \sim \text{Bin}(n, \theta) \rightarrow p(y|\theta) = \binom{n}{y} \theta^{y} (1-\theta)^{n-y}$

Y is the number of success among n Bernoulli trial

Prior:

$\theta \sim \text{Beta}(\alpha, \beta) \rightarrow p(\theta) = \dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$

Posterior :

$P(\theta|y) = \text{Beta}(\theta|\alpha + y, \beta + n - y)$

Posterior predictive distribution :

$$p(\hat{y}|y) = \int_0^1 p(\hat{y}, \theta|y) d\theta = \int_0^1 p(\hat{y}|\theta) p(\theta|y) d\theta$$

$$p(y) = \int_0^1 p(y, \theta) d\theta = \int_0^1 p(y|\theta) p(\theta) d\theta$$

$$= \text{Beta–Binomial}(y|n, \alpha, \beta)$$

Using this result, **the posterior predictive distribution** is

$$p(\hat{y}|y) = \int_0^1 p(\hat{y}, \theta|y) d\theta = \int_0^1 p(\hat{y}|\theta) p(\theta|y) d\theta$$

$$= \text{Beta–Binomial}(\hat{y}|n, \alpha + y, \beta + n - y)$$

$p(\theta) = \text{Beta}(\alpha, \beta)$

$\downarrow$

$p(\theta|y) = \text{Beta}(\alpha + y, \beta + n - y)$

Three Steps in Bayesian Approaches

Step 1: Modeling

setting up a full probability model, a joint probability distribution for all observable and unobservable quantities in a target problem

Step 2: Inferencing

calculate and interpret the appropriate posterior distribution, the conditional probability distribution of the unobserved quantities of interests

Step 3: Checking

Evaluate the fit of the model and the sensitiveness of the assumption in step 1

Conjugacy

From the Bayes rule, $p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{\int_\theta p(y|\theta)p(\theta)d\theta}$

$\mathcal{P}$ $\mathcal{F}$ $\mathcal{P}$

Prior + Likelihood = Posterior

Same form of dist.

$p(\theta|y) \in \mathcal{P}$ for all $p(\cdot\,|\theta) \in \mathcal{F}$ and $p(\cdot) \in \mathcal{P}$

→ $\mathcal{P}$ is conjugate for $\mathcal{F}$

If the posterior is a distribution that is of the same family as our prior
→ the prior is conjugate to the likelihood.

Advantages

- By using conjugate prior, we know the form of the resultant posterior (no math!)
  → we can easily summarize the results using mean, mode, variance, etc.
- Easy to understand the meaning of the prior used in the analysis (insight). For example, Beta prior is just adding pseudo counts to the data.
- Due to the analytical form available, we can carry out the integration

$$p(y) = \int_\theta p(y|\theta)p(\theta)d\theta$$

Conjugate pairs

	Likelihood	Prior	Posterior
Single parameter model	Binomial	Beta	Beta
	Negative Binomial	Beta	Beta
	Geometric	Beta	Beta
	Poisson	Gamma	Gamma
	Exponential	Gamma	Gamma
	Normal (mean unknown)	Normal	Normal
	Normal (variance unknown)	Inverse Gamma	Inverse Gamma
Multi parameters model	Normal (mean and variance unknown)	Normal-Gamma	Normal-Gamma
	Multinomial	Dirichlet	Dirichlet

Conjugate pairs

	Likelihood	Prior	Posterior
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	Multinomial	Dirichlet	Dirichlet

Binomial Likelihood and Beta Prior Distribution (Recap)

Likelihood :

$Y \sim \text{Bin}(n, \theta) \rightarrow p(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$

Prior:

$\theta \sim \text{Beta}(\alpha, \beta) \rightarrow p(\theta) = \dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}$

Prior predictive distribution : $p(y) = \int_\theta p(y, \theta) d\theta = \int_\infty p(y|\theta) p(\theta) d\theta$

$$
\begin{aligned}
p(y) &= \int_0^1 p(y,\theta)d\theta \\
&= \int_0^1 p(y|\theta)p(\theta)d\theta \qquad P(y|\theta) = \text{Bin}(y|n,\theta), \quad p(\theta) = \text{Beta}(\alpha,\beta) \\
&= \int_0^1 \binom{n}{y} \theta^y (1-\theta)^{n-y} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1} d\theta \qquad \binom{n}{y} = \frac{n!}{y!\,(n-y)!} = \frac{\Gamma(n+1)}{\Gamma(y+1)\Gamma(n-y+1)} \\
&= \frac{\Gamma(n+1)}{\Gamma(y+1)\Gamma(n-y+1)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \theta^y (1-\theta)^{n-y} \theta^{\alpha-1}(1-\theta)^{\beta-1} d\theta \\
&= \frac{\Gamma(n+1)\Gamma(\alpha+\beta)}{\Gamma(y+1)\Gamma(n-y+1)\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \theta^{y+\alpha-1}(1-\theta)^{n-y+\beta-1} d\theta \\
&= \frac{\Gamma(n+1)\Gamma(\alpha+b)\Gamma(y+\alpha)\Gamma(n-y+\beta)}{\Gamma(y+1)\Gamma(n-y+1)\Gamma(\alpha)\Gamma(\beta)\Gamma(n+\alpha+\beta)} \int_0^1 \frac{\Gamma(n+\alpha+\beta)}{\Gamma(y+\alpha)\Gamma(n-y+\beta)} \theta^{y+\alpha-1}(1-\theta)^{n-y+\beta-1} d\theta \\
&= \frac{\Gamma(n+1)\Gamma(\alpha+\beta)\Gamma(y+\alpha)\Gamma(n-y+\beta)}{\Gamma(y+1)\Gamma(n-y+1)\Gamma(\alpha)\Gamma(\beta)\Gamma(n+\alpha+\beta)} \int_0^1 \text{Beta}(\theta|y+\alpha, n-y+\beta) d\theta \\
&= \frac{\Gamma(n+1)\Gamma(\alpha+\beta)\Gamma(y+\alpha)\Gamma(n-y+\beta)}{\Gamma(y+1)\Gamma(n-y+1)\Gamma(\alpha)\Gamma(\beta)\Gamma(y+\alpha+\beta)} 1 \\
&= \text{Beta–Binomial}(y|n,\alpha,\beta)
\end{aligned}
$$

The beta-binomial distribution is the binomial distribution in *which the probability of success at each trial is not fixed but random* and follows the beta distribution

$\text{Beta}-\text{bin}(y|n,\alpha,\beta) = \int \text{Bin}(y|n,\theta)\text{Beta}(\theta|\alpha,\beta)d\theta$

Binomial Likelihood and Beta Prior Distribution (Recap)

Likelihood :

$Y \sim \mathrm{Bin}(n, \theta) \rightarrow p(y|\theta) = \binom{n}{y} \theta^{y} (1-\theta)^{n-y}$ Y is the number of successes among n Bernoulli trials

Prior:

$\theta \sim \mathrm{Beta}(\alpha, \beta) \rightarrow p(\theta) = \dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$

Posterior :

$$P(\theta|y) \propto P(y|\theta) p(\theta)$$

$$= \binom{n}{y} \theta^{y} (1-\theta)^{n-y} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$\propto \theta^{y} (1-\theta)^{n-y} \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$= \theta^{\alpha+y-1} (1-\theta)^{\beta+n-y-1}$$

$$= \mathrm{Beta}(\theta | \alpha+y, \beta+n-y)$$

(α is a pseudo-count for the success, while β is a pseudo-count for the failure)

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Posterior predictive distribution :

$$p(\hat{y}|y) = \int_0^1 p(\hat{y}, \theta|y)d\theta = \int_0^1 p(\hat{y}|\theta)p(\theta|y)d\theta$$

$$p(y) = \int_0^1 p(y, \theta)d\theta = \int_0^1 p(y|\theta)p(\theta)d\theta$$

$$= \text{Beta–Binomial}(y|n, \alpha, \beta)$$

Using this result, **the posterior predictive distribution** is

$$p(\hat{y}|y) = \int_0^1 p(\hat{y}, \theta|y)d\theta = \int_0^1 p(\hat{y}|\theta)p(\theta|y)d\theta$$

$$= \text{Beta–Binomial}(\hat{y}|n, \alpha+y, \beta+n-y)$$

$p(\theta) = \text{Beta}(\alpha, \beta)$

$\downarrow$

$p(\theta|y) = \text{Beta}(\alpha+y, \beta+n-y)$

Conjugate pairs

Single parameter model

Multi parameters model

	Likelihood	Prior	Posterior
Single parameter model	Binomial	Beta	Beta
	Negative Binomial	Beta	Beta
	Geometric	Beta	Beta
	Poisson	Gamma	Gamma
	Exponential	Gamma	Gamma
	Normal (mean unknown)	Normal	Normal
	Normal (variance unknown)	Inverse Gamma	Inverse Gamma
Multi parameters model	Normal (mean and variance unknown)	Normal-Gamma	Normal-Gamma
	Multinomial	Dirichlet	Dirichlet

Poisson Likelihood-Gamma Prior

Likelihood :

$$Y_i \sim \text{Poisson}(\lambda) \to p(y_i|\lambda) = \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}$$

Prior:

$$\lambda \sim \text{Gamma}(\alpha, \beta) \to p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

Prior predictive distribution :

$$p(y) = \int_\lambda p(y, \lambda) d\lambda = \int_\lambda p(y|\lambda) p(\lambda) d\lambda$$

$$\begin{aligned}
p(y) &= \int_0^\infty \frac{\lambda^y e^{-\lambda}}{y!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} d\lambda \\
&= \frac{\beta^\alpha}{y!\,\Gamma(\alpha)} \int_0^\infty \lambda^y e^{-\lambda} \lambda^{\alpha-1} e^{-\beta\lambda}\, d\lambda \\
&= \frac{\beta^\alpha}{y!\,\Gamma(\alpha)} \int_0^\infty \lambda^{y+\alpha-1} e^{-(\beta+1)\lambda}\, d\lambda \\
&= \frac{\beta^\alpha}{y!\,\Gamma(\alpha)} \frac{\Gamma(y+\alpha)}{(\beta+1)^{y+\alpha}} \int_0^\infty \underbrace{\frac{(\beta+1)^{y+\alpha}}{\Gamma(y+\alpha)} \lambda^{y+\alpha-1} e^{-(\beta+1)\lambda}\, d\lambda}_{\text{Gamma}(\lambda|\alpha+y, \beta+1)} \\
&= \frac{\beta^\alpha}{y!\,\Gamma(\alpha)} \frac{\Gamma(y+\alpha)}{(\beta+1)^{y+\alpha}} \\
&= \binom{y+\alpha-1}{y} \left(\frac{1}{\beta+1}\right)^y \left(\frac{\beta}{\beta+1}\right)^\alpha = \text{Neg}-\text{bin}(y|\alpha, \beta)
\end{aligned}$$

Negative Binomial distribution ($\text{Neg}-\text{bin}$**) is** a discrete probability distribution of the number of successes y in a sequence of independent and identically distributed Bernoulli trials before a specified (non-random) number of failures (denoted λ) occurs.

$$\text{Neg}-\text{bin}(y|\alpha, \beta) = \int \text{Poisson}(y|\lambda)\text{Gamma}(\lambda|\alpha, \beta) d\lambda$$

Poisson Likelihood-Gamma Prior

Likelihood :

$$Y_i \sim \text{Poisson}(\lambda) \rightarrow p(y_i|\lambda) = \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}$$

$$\text{E}(Y_i) = \lambda$$

Prior:

$$\lambda \sim \text{Gamma}(\alpha, \beta) \rightarrow p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$\text{E}(\lambda) = \frac{\alpha}{\beta}$$

Posterior :

$P(\lambda|y) \propto P(y|\lambda)p(\lambda)$ $\quad y = (y_1, \dots, y_n)$ is a sequence of i.i.d. observation

$$\begin{aligned}
P(\lambda|y) &\propto P(y|\lambda)p(\lambda) \\
&= \left(\prod_{i=1}^{n} P(y_i|\lambda)\right) \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \\
&= \left(\prod_{i=1}^{n} \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}\right) \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \\
&= \frac{\lambda^{\Sigma_i y_i} e^{-n\lambda}}{\prod_i^n y_i!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \\
&\propto \lambda^{n\bar{y}} e^{-n\lambda} \lambda^{\alpha-1} e^{-\beta\lambda} \\
&\propto \lambda^{\alpha+n\bar{y}-1} e^{-(\beta+n)\lambda} \\
&= \text{Gamma}(\alpha + n\bar{y}, \beta + n)
\end{aligned}$$

$$\because \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$\text{E}(\lambda|y) = \frac{\alpha + n\bar{y}}{\beta + n} = \frac{\alpha + \Sigma_i y_i}{\beta + n}$$

(α is a pseudo-count for # of events, while β is a pseudo count for # of observations)

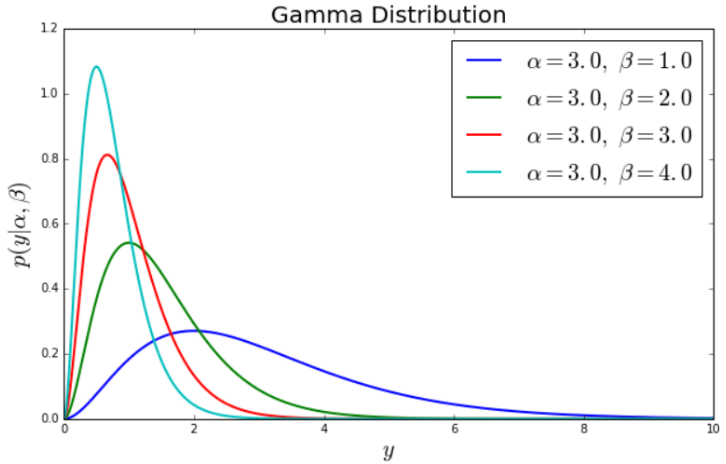
Poisson Likelihood-Gamma Prior

- The posterior mean is

$$
\begin{aligned}
\mathbb{E}[\lambda|y] &= \frac{\alpha + \sum_i y_i}{\beta + n} \\
&= \frac{\alpha}{\beta + n} + \frac{\sum_i y_i}{\beta + n} \\
&= \frac{\beta}{\beta + n}\left(\frac{\alpha}{\beta}\right) + \frac{n}{\beta + n}\left(\frac{\sum_i y_i}{n}\right) \\
&= \frac{\beta}{\beta + n}\mathbb{E}[\lambda] + \frac{n}{\beta + n}\hat{\theta}_{\mathrm{ML}}
\end{aligned}
$$

- Again, the data get weighted more heavily as $n \to \infty$

β control strength of prior



Poisson Likelihood-Gamma Prior

Example 2.1

Counts the number of Pokemons in each district of SF. The numbers are

$$14 \quad 13 \quad 7 \quad 10 \quad 15 \quad 15 \quad 2 \quad 13 \quad 13 \quad 11 \quad 10 \quad 13 \quad 5 \quad 13 \quad 9 \quad 12 \quad 9 \quad 12 \quad 8 \quad 7$$

- It is assumed that the number are independent and drawn from a Poisson distribution with mean λ

$$Y_i \sim \text{Poisson}(\lambda) \to p(y_i|\lambda) = \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}$$

- The prior distribution for λ is a Gamma distribution with mean 20 and standard deviation 10

$$\lambda \sim \text{Gamma}(\alpha, \beta) \to p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$\mathbb{E}[\lambda] = \frac{\alpha}{\beta} = 20, \qquad \text{Var}(\lambda) = \frac{\alpha}{\beta^2} = 10^2 \quad \to \quad \alpha = 4, \beta = 0.2$$

- The posterior is

$$p(\lambda|y) = \text{Gamma}(4 + 211, 0.2 + 20) \qquad P(\lambda|y) = \text{Gamma}(\alpha + n\bar{y}, \beta + n)$$

$$\mathbb{E}[\lambda|y] = \frac{215}{20.2} = 10.64, \qquad \text{Var}(\lambda) = \frac{215}{20.2^2} = 0.5269$$

Poisson Likelihood-Gamma Prior

Likelihood :

$Y_i \sim \text{Poisson}(\lambda) \rightarrow p(y_i|\lambda) = \dfrac{\lambda^{y_i} e^{-\lambda}}{y_i!}$

Prior:

$\lambda \sim \text{Gamma}(\alpha, \beta) \rightarrow p(\lambda) = \dfrac{\beta^{\alpha}}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$

Posterior :

$P(\lambda|y) = \text{Gamma}(\alpha + n\bar{y}, \beta + n)$

$y = (y_1, \ldots, y_n)$ is a sequence of i.i.d. observations

Posterior predictive distribution (Using Bayes' rule)

$$p(\hat{y}|y) = \int_{\lambda} p(\hat{y}, \lambda|y)d\lambda = \int_{\lambda} p(\hat{y}|\lambda)p(\lambda|\text{y})d\lambda$$

$$p(y) = \int_0^{\infty} p(y, \lambda)d\lambda = \int_0^{\infty} p(y|\lambda)p(\lambda)d\lambda$$

$$= NB(y|\alpha, \beta) = \binom{y + \alpha - 1}{y} \left(\frac{1}{\beta + 1}\right)^{y} \left(\frac{\beta}{\beta + 1}\right)^{\alpha}$$

$p(\theta) = \text{Gamma}(\alpha, \beta)$

$p(\theta|y) = \text{Gamma}(\alpha + n\bar{y}, \beta + n)$

Using this result, **the posterior predictive distribution** is

$$p(\hat{y}|y) = \int_0^{\infty} p(\hat{y}, \theta|\text{y})d\lambda = \int_0^{\infty} p(\hat{y}|\theta)p(\theta|\text{y})d\lambda$$

$$= NB(\hat{y}|\alpha + n\bar{y}, \beta + n) = \binom{\hat{y} + \alpha + n\bar{y} - 1}{\hat{y}} \left(\frac{1}{\beta + n + 1}\right)^{\hat{y}} \left(\frac{\beta + n}{\beta + n + 1}\right)^{\alpha + n\bar{y}}$$

Conjugate pairs

	Likelihood	Prior	Posterior
Single parameter model	Binomial	Beta	Beta
	Negative Binomial	Beta	Beta
	Geometric	Beta	Beta
	Poisson	Gamma	Gamma
	Exponential	Gamma	Gamma
	Normal (mean unknown)	Normal	Normal
	Normal (variance unknown)	Inverse Gamma	Inverse Gamma
Multi parameters model	Normal (mean and variance unknown)	Normal-Gamma	Normal-Gamma
	Multinomial	Dirichlet	Dirichlet

Normal Likelihood-Normal Prior (unknown mean and known variance)

Likelihood :

$$Y_i \sim \mathrm{N}(\theta, \sigma_Y^2) \rightarrow p(y_i|\theta, \sigma_Y^2) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left(-\frac{(y_i - \theta)^2}{2\sigma_Y^2}\right)$$

Note that $\theta = \mu_Y$, (σ_Y^2 is known)

Prior:

$$\theta \sim \mathrm{N}(\mu_0, \tau_0^2) \rightarrow p(\theta) = \frac{1}{\sqrt{2\pi\tau_o^2}} \exp\left(-\frac{(\theta - \mu_0)^2}{2\tau_o^2}\right)$$

Prior predictive distribution :

Before the data y are considered, the distribution of the unknown but observable y is

$$p(y) = \int_\theta p(y, \theta) d\theta = \int_{-\infty}^{\infty} p(y|\theta) p(\theta) d\theta$$

(Without integration)

$$\mathbb{E}[y] = \mathbb{E}\big[\mathbb{E}[y|\theta]\big] = \mathbb{E}[\theta] = \mu_0$$

$\because \mathbb{E}(y|\theta) = \theta$

$$\begin{aligned} \mathrm{Var}(y) &= \mathbb{E}[\mathrm{Var}(y|\theta)] + \mathrm{Var}(\mathbb{E}[y|\theta]) \\ &= \mathbb{E}[\sigma_Y^2] + \mathrm{Var}(\theta) \\ &= \sigma_Y^2 + \tau_0^2 \end{aligned}$$

$\because \mathrm{Var}(y|\theta) = \sigma_Y^2$

$$p(y) = \mathrm{N}(y|\mu_0, \sigma_Y^2 + \tau_0^2)$$

Normal Likelihood-Normal Prior (unknown mean and known variance)

Likelihood :

$$Y_i \sim \mathrm{N}(\theta, \sigma_Y^2) \rightarrow p(y_i|\theta, \sigma_Y^2) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left(-\frac{(y_i-\theta)^2}{2\sigma_Y^2}\right)$$

Note that $\theta = \mu_Y$, (σ_Y^2 is known)

Prior:

$$\theta \sim \mathrm{N}(\mu_0, \tau_0^2) \rightarrow p(\theta) = \frac{1}{\sqrt{2\pi\tau_o^2}} \exp\left(-\frac{(\theta-\mu_0)^2}{2\tau_o^2}\right)$$

Posterior :

$y = (y_1, \ldots, y_n)$ is a sequence of i.i.d. observations

$$\begin{aligned}
p(\theta|y) &\propto p(y|\theta, \sigma_Y^2)p(\theta) \\
&= \left(\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i-\theta)^2}{2\sigma_Y^2}\right)\right) \frac{1}{\sqrt{2\pi\tau_o^2}} \exp\left(-\frac{(\theta-\mu_0)^2}{2\tau_o^2}\right) \\
&\propto \exp\left(-\sum_{i=1}^{n} \frac{(y_i-\theta)^2}{2\sigma_Y^2} + \frac{(\theta-\mu_0)^2}{2\tau_0^2}\right) \\
&= \exp\left[-\frac{1}{2}\left(\sum_{i=1}^{n} \frac{(y_i-\theta)^2}{\sigma_Y^2} + \frac{(\theta-\mu_0)^2}{\tau_0^2}\right)\right] \\
&= \exp\left[-\frac{1}{2\sigma_Y^2\tau_0^2}\left(\tau_0^2 \sum_{i=1}^{n} (y_i-\theta)^2 + \sigma_Y^2(\theta-\mu_0)^2\right)\right] \\
&= \exp\left[-\frac{1}{2\sigma_Y^2\tau_0^2}\left(\tau_0^2 \sum_{i=1}^{n} (y_i^2 - 2\theta y_i + \theta^2) + \sigma_Y^2(\theta^2 - 2\theta\mu_0 + \mu_0^2)\right)\right]
\end{aligned}$$

Normal Likelihood-Normal Prior (unknown mean and known variance)

Posterior :

$$p(\theta|y) \propto \exp\left[-\frac{1}{2\sigma_Y^2\tau_0^2}\left(\tau_0^2\sum_{i=1}^{n}\left(y_i^2-2\theta y_i+\theta^2\right)+\sigma_Y^2(\theta^2-2\theta\mu_0+\mu_0^2)\right)\right]$$

$$= \exp\left[-\frac{1}{2\sigma_Y^2\tau_0^2}\left(\tau_0^2\sum_{i=1}^{n}y_i^2-2\tau_0^2\theta n\bar{y}+\tau_0^2 n\theta^2+\sigma_Y^2\theta^2-2\sigma_Y^2\theta\mu_0+\sigma_Y^2\mu_0^2\right)\right]$$

$$\left(\because \bar{y}=\frac{1}{n}\sum_{i=1}^{n}y_i\right)$$

$$= \exp\left[-\frac{1}{2\sigma_Y^2\tau_0^2}\left(\tau_0^2\sum_{i=1}^{n}y_i^2-2\tau_0^2\theta n\bar{y}+\tau_0^2 n\theta^2+\sigma_Y^2\theta^2-2\sigma_Y^2\theta\mu_0+\sigma_Y^2\mu_0^2\right)\right]$$

$$\propto \exp\left[-\frac{1}{2\sigma_Y^2\tau_0^2}\left(\theta^2(\sigma_Y^2+n\tau_0^2)-2\theta(\mu_0\sigma_Y^2+n\bar{y}\tau_0^2)+\mathrm{const}\right)\right]$$

$$= \exp\left[-\frac{1}{2}\left(\theta^2\left(\frac{\sigma_Y^2+n\tau_0^2}{\sigma_Y^2\tau_0^2}\right)-2\theta\left(\frac{\mu_0\sigma_Y^2+n\bar{y}\tau_0^2}{\sigma^2\tau_0^2}\right)+\mathrm{const}'\right)\right]$$

$$= \exp\left[-\frac{1}{2}\left(\theta^2\left(\frac{1}{\tau_0^2}+\frac{n}{\sigma_Y^2}\right)-2\theta\left(\frac{\mu_0}{\tau_0^2}+\frac{n\bar{y}}{\sigma_Y^2}\right)+\mathrm{const}'\right)\right]$$

$$= \exp\left[-\frac{1}{2}\left(\frac{1}{\tau_0^2}+\frac{n}{\sigma_Y^2}\right)\left(\theta^2-2\theta\left(\frac{\frac{\mu_0}{\tau_0^2}+\frac{n\bar{y}}{\sigma_Y^2}}{\frac{1}{\tau_0^2}+\frac{n}{\sigma_Y^2}}\right)+\mathrm{const}'\right)\right]$$

$$= \exp\left[-\frac{1}{2}\left(\frac{1}{\tau_0^2}+\frac{n}{\sigma_Y^2}\right)\left(\theta-\left(\frac{\frac{\mu_0}{\tau_0^2}+\frac{n\bar{y}}{\sigma_Y^2}}{\frac{1}{\tau_0^2}+\frac{n}{\sigma_Y^2}}\right)\right)^2\right]$$

Normal Likelihood-Normal Prior (unknown mean and known variance)

Likelihood :

$$Y_i \sim \mathrm{N}(\theta, \sigma_Y^2) \rightarrow p(y_i|\theta, \sigma_Y^2) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left(-\frac{(y_i - \theta)^2}{2\sigma_Y^2}\right)$$

Prior:

$$\theta \sim \mathrm{N}(\mu_0, \tau_0^2) \rightarrow p(\theta) = \frac{1}{\sqrt{2\pi\tau_o^2}} \exp\left(-\frac{(\theta - \mu_0)^2}{2\tau_o^2}\right)$$

Posterior on $\theta = \mu_Y$

$$p(\theta|y) = \mathrm{N}\left(\theta \left| \frac{\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma_Y^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma_Y^2}}, \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma_Y^2}\right)^{-1} \right.\right)$$

- Posterior mean μ_1 :

$$\mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma_Y^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma_Y^2}} = \frac{\sigma_Y^2 \mu_0}{\sigma_Y^2 + n\tau_0^2} + \frac{n\tau_0^2 \bar{y}}{\sigma_Y^2 + n\tau_0^2}$$

μ_0: Prior mean

$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$: Data mean

  - $\tau_0^2 \downarrow \Rightarrow$ Prior mean μ_0 becomes accurate and influence more to μ_1
  - $\sigma_Y^2 \downarrow \Rightarrow$ the data become precise, making $\bar{y}$ stronger
  - $n \uparrow \Rightarrow \bar{y}$ stronger

- Posterior variance τ_1^2 :

$$\tau_1^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma_Y^2}\right)^{-1}$$

- Posterior precision :

$$\frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma_Y^2}$$

$\frac{1}{\tau_0^2}$: Prior precision

$\frac{n}{\sigma_Y^2}$: Data precision

Normal Likelihood-Normal Prior (unknown mean and known variance)

Likelihood :

$$Y_i \sim \mathrm{N}(\theta, \sigma_Y^2) \rightarrow p(y_i|\theta, \sigma_Y^2) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left(-\frac{(y_i-\theta)^2}{2\sigma_Y^2}\right)$$

Note that $\theta = \mu_Y$, (σ_Y^2 is known)

Posterior : $\theta = \mu_Y$

$$p(\theta|y) = \mathrm{N}(\theta|\mu_1, \tau_1^2)$$

$$\mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma_Y^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma_Y^2}} \qquad \tau_1^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma_Y^2}\right)^{-1}$$

$$E[\theta|y] = \mu_1 \qquad \mathrm{var}(\theta|y) = \tau_1^2$$

Posterior predictive distribution

$$p(\hat{y}|y) = \int_\theta p(\hat{y}, \theta|y)d\theta = \int_\theta p(\hat{y}|\theta, y)p(\theta|y)d\theta = \int_\theta p(\hat{y}|\theta)p(\theta|y)d\theta$$

(Without integration)

$$\mathbb{E}[\hat{y}|y] = \mathbb{E}[\mathbb{E}[\hat{y}|\theta, y]|y] = \mathbb{E}[\theta|y] = \mu_1 \qquad \because \mathbb{E}[\hat{y}|\theta, y] = \mathbb{E}[\hat{y}|\theta] = \theta$$

$$\begin{aligned}\mathrm{Var}(\hat{y}|y) &= \mathbb{E}[\mathrm{Var}(\hat{y}|\theta, y)|y] + \mathrm{Var}(\mathbb{E}[\hat{y}|\theta, y]|y) \qquad \because \mathrm{Var}(\hat{y}|\theta, y) = \sigma_Y^2 \\ &= \mathbb{E}[\sigma_Y^2|y] + \mathrm{Var}(\theta|y) \\ &= \sigma_Y^2 + \tau_1^2\end{aligned}$$

$$p(\hat{y}|y) = \mathrm{N}(\hat{y}|\mu_1, \sigma_Y^2 + \tau_1^2)$$

Normal Likelihood-Normal Prior (unknown mean and known variance)

$$\mathbb{E}[y] = \mathbb{E}[\mathbb{E}(y|\theta)] = \mathbb{E}[\theta] = \mu_0 \qquad \because\ \mathbb{E}[y|\theta] = \theta$$

$$\begin{aligned}\mathrm{Var}(y) &= \mathbb{E}[\mathrm{Var}(y|\theta)] + \mathrm{Var}(\mathbb{E}[y|\theta]) \qquad \because\ \mathrm{Var}(y|\theta) = \sigma_Y^2 \\ &= \mathbb{E}[\sigma_Y^2] + \mathrm{Var}(\theta) \\ &= \sigma_Y^2 + \tau_0^2\end{aligned}$$

$$p(y) = \mathrm{N}(y|\mu_0, \sigma_Y^2 + \tau_0^2)$$

$$\mathbb{E}[\hat{y}|y] = \mathbb{E}[\mathbb{E}[\hat{y}|\theta, y]|y] = \mathbb{E}[\theta|y] = \mu_1 \qquad \because\ \mathbb{E}[\hat{y}|\theta, y] = \mathbb{E}[\hat{y}|\theta] = \theta$$

$$\begin{aligned}\mathrm{Var}(\hat{y}|y) &= \mathbb{E}[\mathrm{Var}(\hat{y}|\theta, y)|y] + \mathrm{Var}(\mathbb{E}[\hat{y}|\theta, y]|y) \qquad \because\ \mathrm{Var}(\hat{y}|\theta, y) = \sigma_Y^2 \\ &= \mathbb{E}[\sigma_Y^2|y] + \mathrm{Var}(\theta|y) \\ &= \sigma_Y^2 + \tau_1^2\end{aligned}$$

$$p(\hat{y}|y) = \mathrm{N}(\hat{y}|\mu_1, \sigma_Y^2 + \tau_1^2)$$

Conjugate pairs

	Likelihood	Prior	Posterior
Single parameter model	Binomial	Beta	Beta
	Negative Binomial	Beta	Beta
	Geometric	Beta	Beta
	Poisson	Gamma	Gamma
	Exponential	Gamma	Gamma
	Normal (mean unknown)	Normal	Normal
	Normal (variance unknown)	Inverse Gamma	Inverse Gamma
Multi parameters model	Normal (mean and variance unknown)	Normal-Gamma	Normal-Gamma
	Multinomial	Dirichlet	Dirichlet

Multinomial Likelihood-Dirichlet Prior

Likelihood :

$$p(y|\theta) = \text{Multin}(y|n, \theta_1, \ldots, \theta_k) = \binom{n}{y_1\ y_2 \cdots y_k} {\theta_1}^{y_1} \cdots {\theta_k}^{y_k}$$

$$= \frac{\Gamma(n+1)}{\prod_{j=1}^{k} \Gamma(y_j+1)} \prod_{j=1}^{k} \theta_j^{y_j}$$

$y = (y_1, \ldots, y_j, \ldots, y_k)$

$y_j \in \{0,1,\ldots,n\}, \sum_{j=1}^{k} y_j = n$

Prior:

$$\theta \sim \text{Dirichlet}(\alpha_1, \ldots, \alpha_k) \rightarrow p(\theta) = \frac{\Gamma(\alpha_1 + \cdots + \alpha_k)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_k)} \theta_1^{\alpha_1 - 1} \cdots \theta_k^{\alpha_k - 1}$$

$$= \frac{\Gamma(\sum_{j=1}^{k} \alpha_j)}{\prod_{j=1}^{k} \Gamma(\alpha_k)} \theta_1^{\alpha_1 - 1} \cdots \theta_k^{\alpha_k - 1}$$

$\sum_{j=1}^{k} \theta_j = 1$

Posterior :

$$p(\theta|y) \propto p(y|\theta\,)p(\theta)$$

$$= \frac{\Gamma(n+1)}{\prod_{j=1}^{k} \Gamma(y_j+1)} \prod_{j=1}^{K} \theta_j^{y_j} \frac{\Gamma(\sum_{j=1}^{k} \alpha_j)}{\prod_{j=1}^{k} \Gamma(\alpha_j)} \theta_1^{\alpha_1 - 1} \cdots \theta_k^{\alpha_k - 1}$$

$$\propto \prod_{j=1}^{k} \theta_j^{y_j} \prod_{j=1}^{k} \theta_j^{\alpha_j - 1}$$

$$\propto \prod_{j=1}^{k} \theta_j^{\alpha_j + y_j - 1}$$

$$= \text{Dirichlet}(\alpha_1 + y_1, \ldots, \alpha_k + y_k)$$

Multinomial Likelihood-Dirichlet Prior

Posterior predictive distribution

$$p(\hat{y}|y) = \int_\theta p(\hat{y},\theta|y)d\theta = \int_\theta p(\hat{y}|\theta,y)p(\theta|y)d\theta = \int_\theta p(\hat{y}|\theta)p(\theta|y)d\theta$$

$$\begin{aligned}
p(\hat{y}|y) &= \int_\theta p(\hat{y}|\theta_1,\dots,\theta_k)p(\theta_1,\dots,\theta_k|y)d\theta \\
&= \int_\theta \frac{\Gamma(n+1)}{\prod_{j=1}^k \Gamma(y_j+1)} \prod_{j=1}^k \theta_j^{y_j} \frac{\Gamma(\sum_{j=1}^k \alpha_j)}{\prod_{j=1}^k \Gamma(\alpha_j)} \prod_{j=1}^k \theta_j^{\alpha_j-1}\, d\theta \\
&= \frac{\Gamma(n+1)}{\prod_{j=1}^K \Gamma(y_j+1)} \frac{\Gamma(\sum_{j=1}^K \alpha_j)}{\prod_{j=1}^K \Gamma(\alpha_j)} \int_\theta \prod_{j=1}^K \theta_j^{y_j+\alpha_j-1}\, d\theta \\
&= \frac{\Gamma(n+1)}{\prod_{j=1}^K \Gamma(y_j+1)} \frac{\Gamma(\sum_{j=1}^K \alpha_j)}{\prod_{j=1}^K \Gamma(\alpha_j)} \frac{\prod_{j=1}^K \Gamma(y_j+\alpha_j)}{\Gamma(n+\sum_{j=1}^K \alpha_j)} \int_\theta \frac{\Gamma(n+\sum_{j=1}^K \alpha_j)}{\prod_{j=1}^K \Gamma(y_j+\alpha_j)} \prod_{j=1}^K \theta_j^{y_j+\alpha_j-1}\, d\theta \\
&= \frac{\Gamma(n+1)}{\prod_{j=1}^K \Gamma(y_j+1)} \frac{\Gamma(\sum_{j=1}^K \alpha_j)}{\prod_{j=1}^K \Gamma(\alpha_j)} \frac{\prod_{j=1}^K \Gamma(y_j+\alpha_j)}{\Gamma(n+\sum_{j=1}^K \alpha_j)}
\end{aligned}$$

Bayesian Regression

Supervised learning

Data

Training data set D

$D = \{(x_i, y_i); i = 1, \cdots, m\}$

$x_i = (x_{i1}, \cdots, x_{in})$

$y_i \in \mathbb{R}$ or $y_i \in \{1, \cdots, N\}$

m input Feature vectors

x_1	x_2	$\cdots$	x_n	y
x_{11}	x_{12}	$\cdots$	x_{1n}	y_1
x_{21}	x_{22}	$\cdots$	x_{2n}	y_2
$\vdots$	$\vdots$		$\vdots$	$\vdots$
x_{m1}	x_{m2}	$\cdots$	x_{mn}	y_m

m outputs

model

Functional form $f(x;\theta)$ Is usually given

Learning Algorithm

Using training data set, a learning algorithm finds the best hypothesis function $h(x)$ that **is believed to accurately predict** the output y for a given query input x

prediction

θ^*

Query input x^* → Hypothesis $f(x;\theta^*)$ → Predicted output y^*

Input feature vector

$x^* = (x_{i1}^*, x_{i2}^*, \cdots, x_{in}^*)_{i \in [m]}$

If $y^* \in \mathbb{R}$: **Regression**

If $y^* \in \{1, \cdots, N\}$: **Classification**

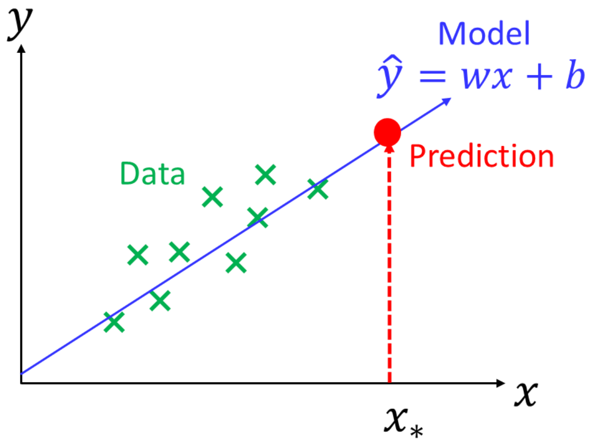
Problem solving

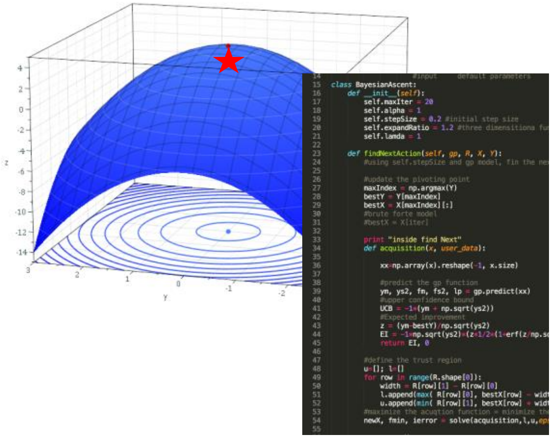
Estimate a housing price given historical house sales data

Modeling

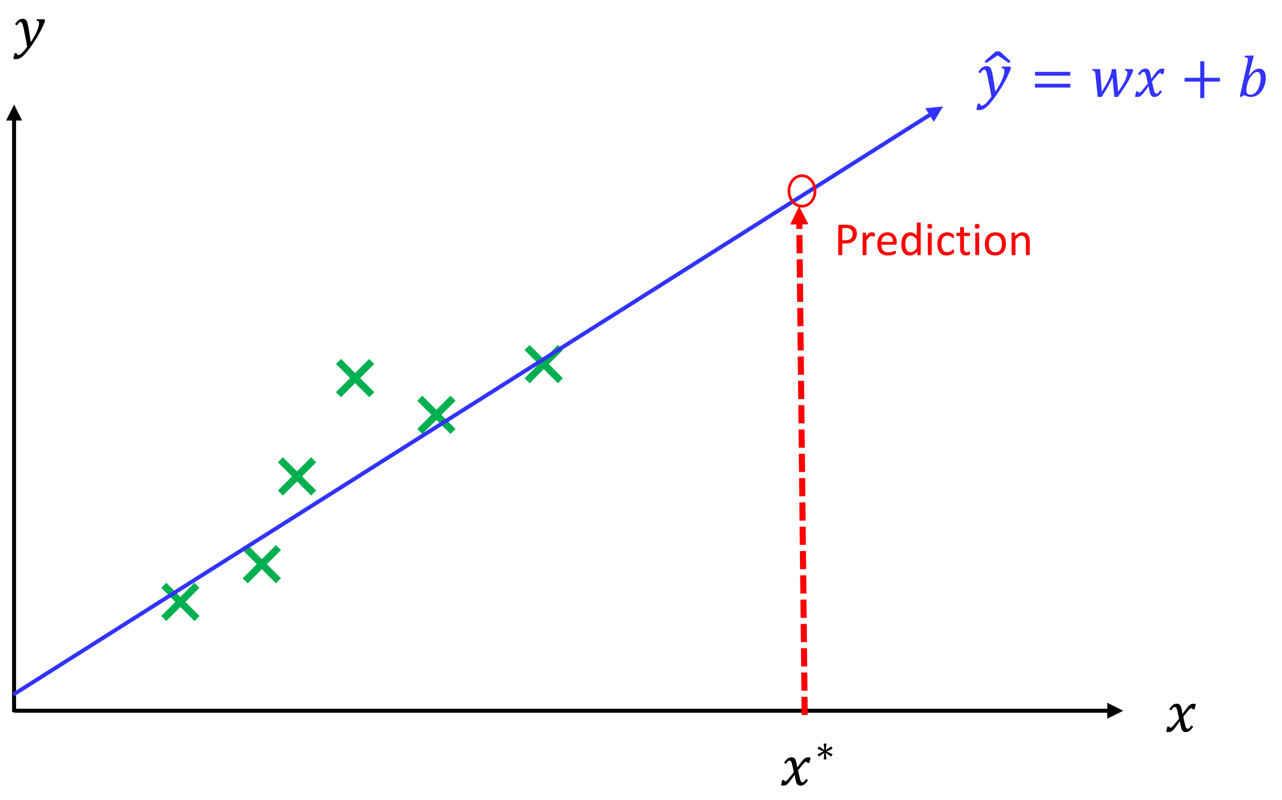
Solving

Real-world task → **Formal task (model)** → **Algorithm (program)**





Linear regression



- **Data**: $(x_1, y_1), \cdots, (x_m, y_m)$
- **Model**: Linear model $\hat{y} = wx + b$ ($y = w^{\mathrm{T}}x + b$ for multi-dimensional)
  - Learning: What are the optimum w and b?
- **Prediction**: What is $\hat{y}^* = wx^* + b$

Load Map

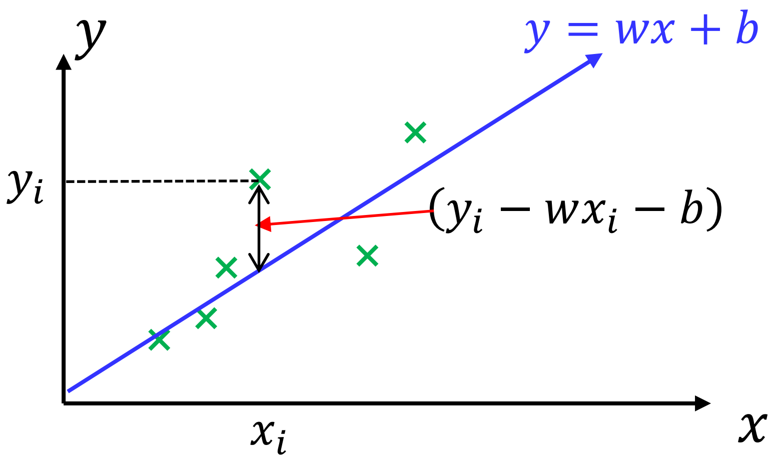
We are going to formulate linear regression using various approaches
(to show the connection between different approaches)

- Optimization Approach (Normal Equation)
- Maximum Likelihood Estimation (MLE) Approach
- Maximum A Posteriori Estimation (MAP) Approach
- Full Bayesian approach
  - ✓ Analytical approach
  - ✓ Sampling approach
- Regularization regression (Ridge and Lasso)
  - ✓ Optimization view
  - ✓ Bayesian view

Learning as optimization

- Define an objective (cost) function

$$J(w,b) = \sum_{i=1}^{m} (y_i - wx_i - b)^2$$



- Minimize the error function with respect to w and b

$$\frac{\partial J(w,b)}{\partial w} = -2\sum_{i=1}^{m} x_i(y_i - wx_i - b) = 0 \;\rightarrow w^* = \frac{\sum_{i=1}^{m}(y_i - b)x_i}{\sum_{i=1}^{n} x_i^2}$$

$$\frac{\partial J(w,b)}{\partial b} = -2\sum_{i=1}^{m} (y_i - wx_i - b) = 0 \;\rightarrow b^* = \frac{\sum_{i=1}^{m}(y_i - wx_i)}{n}$$

Learning as optimization

- Notation for general cases $x_i \in \mathbb{R}^n$
- A linear regression model

$$\hat{y}_i = w_0 + w_1 x_{i1} + \cdots + w_n x_{in}$$

where $w = (w_0, w_1, \cdots, w_n)^{\mathrm{T}}$, and $x_i = (x_{i1}, x_{i2}, \cdots, x_{in})^{\mathrm{T}}$.

- If we introduce $x_{i0} = 1$,

$$\hat{y}_i = w^{\mathrm{T}} x_i$$

where $(w_0, w_1, \cdots, w_n)^{\mathrm{T}}$, and $x_i = (x_{i0}, x_{i1}, x_{i2}, \cdots, x_{in})^{\mathrm{T}}$

1

- In a Matrix form

$$\begin{pmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_m \end{pmatrix} = \begin{pmatrix} -x_1^{\mathrm{T}}- \\ \vdots \\ -x_m^{\mathrm{T}}- \end{pmatrix} \begin{pmatrix} w_0 \\ \vdots \\ w_n \end{pmatrix} = \begin{pmatrix} x_1^{\mathrm{T}} w \\ \vdots \\ x_m^{\mathrm{T}} w \end{pmatrix} \rightarrow \hat{y} = \mathbf{X} w$$

where m is the number of data points, $\hat{y} = (\hat{y}_1, \cdots, \hat{y}_m)^{\mathrm{T}}$, and $\mathbf{X} = \left(x_1^{\mathrm{T}}, \cdots, x_m^{\mathrm{T}}\right)^{\mathrm{T}}$.

Learning as optimization (Normal Equation)

- The cost function for the optimization can be defined as:

$$J(w) = \frac{1}{2}\sum_{i=1}^{m}\left(x_i^{\mathrm{T}}w - y_i\right)^2 = \frac{1}{2}\|y - w\mathbf{X}\|_2^2 = \frac{1}{2}(\mathbf{X}w - y)^T(\mathbf{X}w - y)$$

$$\left(\sum_{i=1}^{m} z_i^2\right)^{\frac{1}{2}} \quad = \quad \|z\|_2 \quad = \quad \sqrt{z^{\mathrm{T}}z}$$

- The optimum parameters $\widehat{w}$ can be computed as one minimizing the cost function:

$$\widehat{w} = \underset{w}{\operatorname{argmin}}\, J(w) = \underset{w}{\operatorname{argmin}}\frac{1}{2}\|y - w\mathbf{X}\|_2^2$$

- For reference, other norm are summarized here:

$$\|z\|_1 = \sum_{i=1}^{n}|z_i|, \qquad \|z\|_p = \left(\sum_{i=1}^{n}|z_i|^p\right)^{\frac{1}{p}}, \qquad \|z\|_\infty = \max_i |z_i|$$

Learning as optimization (Normal Equation)

- Linear Algebra Approach for finding the optimum parameters:

$$\hat{w} = \underset{w}{\operatorname{argmin}}\, J(w) = \underset{w}{\operatorname{argmin}} \frac{1}{2} \|y - w\mathbf{X}\|_2^2$$

$$\text{Optimality condition: } \nabla_w J(w) = 0 \text{ at } \hat{w}$$

$$\begin{aligned}
\nabla_w J(w) &= \nabla_w \frac{1}{2} (\mathbf{X}w - y)^{\mathrm{T}} (\mathbf{X}w - y) \\
&= \frac{1}{2} \nabla_w \left(w^{\mathrm{T}} \mathbf{X}^{\mathrm{T}} \mathbf{X} w - w^{\mathrm{T}} \mathbf{X}^{\mathrm{T}} y - y^{\mathrm{T}} \mathbf{X} w + y^{\mathrm{T}} y \right) \\
&= \frac{1}{2} \nabla_w \mathrm{tr}\left(w^{\mathrm{T}} \mathbf{X}^{\mathrm{T}} X w - w^{\mathrm{T}} \mathbf{X}^{\mathrm{T}} y - y^{\mathrm{T}} \mathbf{X} w + y^{\mathrm{T}} y \right) \\
&= \frac{1}{2} \nabla_w \left(\mathrm{tr}\left(w^{\mathrm{T}} \mathbf{X}^{\mathrm{T}} X w \right) - 2\mathrm{tr}\left(y^{\mathrm{T}} \mathbf{X} w \right) \right) \\
&= \frac{1}{2} \left(\mathbf{X}^{\mathrm{T}} \mathbf{X} w + \mathbf{X}^{\mathrm{T}} \mathbf{X} w - 2\mathbf{X}^{\mathrm{T}} y \right) = \mathbf{X}^{\mathrm{T}} \mathbf{X} w - \mathbf{X}^{\mathrm{T}} y
\end{aligned}$$

$$\begin{aligned}
\nabla_w J(w) &= \mathbf{X}^{\mathrm{T}} \mathbf{X} w - \mathbf{X}^{\mathrm{T}} y = 0 \\
&\rightarrow \mathbf{X}^{\mathrm{T}} \mathbf{X} w = \mathbf{X}^{\mathrm{T}} y \\
&\rightarrow \hat{w} = \left(\mathbf{X}^{\mathrm{T}} \mathbf{X} \right)^{-1} \mathbf{X}^{\mathrm{T}} y
\end{aligned}$$

Probabilistic view on linear regression

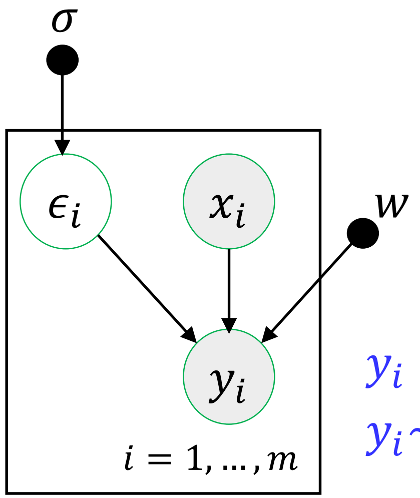
- Assume there is uncertainty in the predicted value:

$$y_i = w^{\mathrm{T}} x_i + \epsilon_i \quad \text{with} \quad \epsilon_i \sim \mathrm{N}(0, \sigma^2)$$

- Then the probabilistic model on output y_i can be represented as

$$y_i \sim \mathrm{N}(w^T x_i, \sigma^2) \quad \text{or} \quad p\left(y_i | w^{\mathrm{T}} x_i, \sigma\right) = \mathrm{N}\left(y_i | w^{\mathrm{T}} x_i, \sigma^2\right)$$

$\epsilon_i \sim \mathrm{N}(0, \sigma^2)$



$y_i = w^{\mathrm{T}} x_i + \epsilon_i$

$y_i \sim \mathrm{N}\left(w^{\mathrm{T}} x_i, \sigma^2\right)$

An error ϵ_i is independently identically distributed (i.i.d assumption)

- The likelihood of the data is defined as

$$p(y | \mathbf{X}, w, \sigma) = \prod_{i=1}^{m} \mathrm{N}(y_i | w^T x_i, \sigma^2) = \prod_{i=1}^{m} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\left(y_i - w^{\mathrm{T}} x_i\right)^2}{2\sigma^2}\right)$$

Learning as probabilistic model (MLE approach)

- The log likelihood is

$$
\begin{aligned}
L(w,\sigma) &= \log p(y|\mathbf{X},w,\sigma) \\
&= \log \prod_{i=1}^{m} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\left(y_i - w^{\mathrm{T}}x_i\right)^2}{2\sigma^2}\right) \\
&= \log \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^m \exp\left(-\sum_{i=1}^{m} \frac{\left(y_i - w^{\mathrm{T}}x_i\right)^2}{2\sigma^2}\right) \\
&= m \log \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{1}{\sigma^2}\frac{1}{2}\sum_{i=1}^{m}\left(y_i - w^{\mathrm{T}}x_i\right)^2 \\
&= m \log \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{1}{\sigma^2} J(w)
\end{aligned}
$$

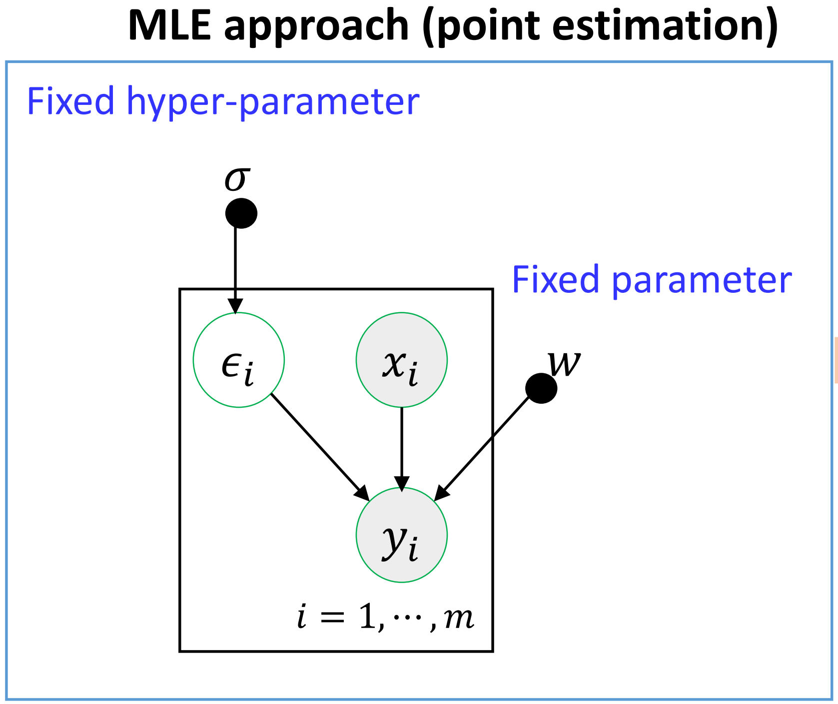
- The optimum parameters is determined by maximizing log likelihood

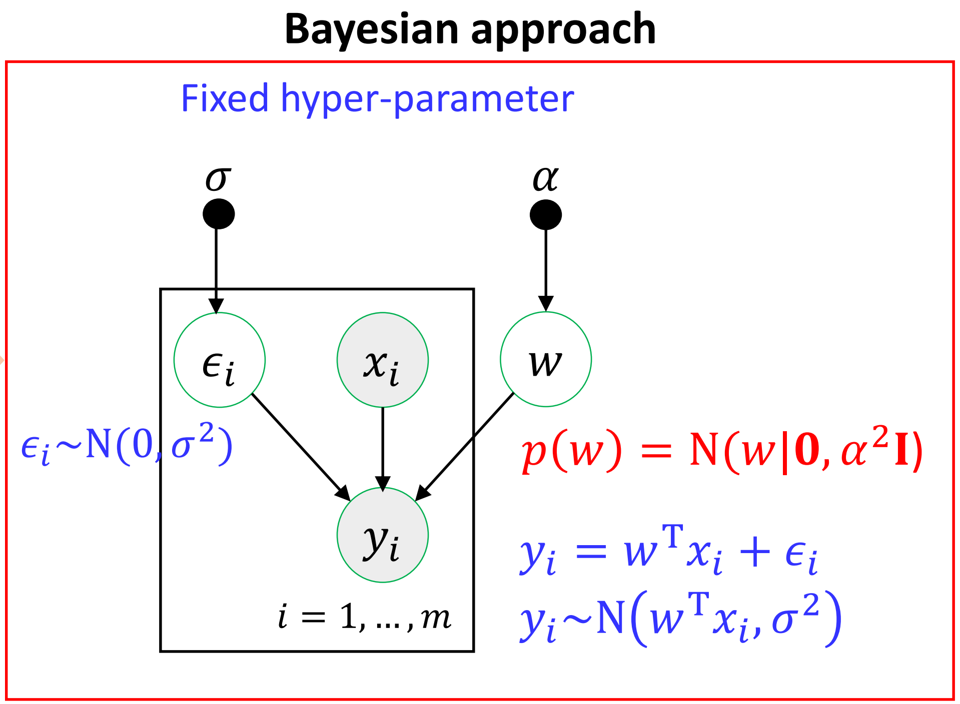
$$
(w^*,\sigma) = \max_{(w,\sigma)} L(w,\sigma) = \max_{(w,\sigma)} \log p(y|\mathbf{X},w,\sigma)
$$

Maximizing the log likelihood $\log p(y|\mathbf{X},w,\sigma)$ with respect to w

= Minimizing the square error sum $J(w)$

Learning as probabilistic model (Bayesian approach)





- Consider the parameter w as random variables (represented as a distribution)
- (Assume σ is known for simple derivation, but can be random, too)
- Find the distribution on parameter w

$$p(w|y, \mathbf{X}) = \frac{p(y|\mathbf{X}, w)p(w|\mathbf{X})}{\int_w p(y|\mathbf{X}, w)p(w|\mathbf{X})dw} \rightarrow p(w|y) = \frac{p(y|w)p(w)}{\int_w p(y|w)p(w)dw}$$

We will assume $\mathbf{X}$ is fixed for the data y

Learning as probabilistic model (Bayesian approach)

- Multivariate Gaussian likelihood is

$$p(y|w) = \prod_{i=1}^{m} p(y_i|x_i, w)$$

$$= \frac{1}{(2\pi\sigma^2)^{\frac{m}{2}}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{m}\left(y_i - w^{\mathrm{T}}x_i\right)^2\right)$$

m = # of data points

- Multivariate Gaussian prior on parameter w

$$p(w) = \mathrm{N}(w|\mathbf{0}, \alpha^2\mathbf{I})$$

$$p(w) = \frac{1}{(2\pi\alpha^2)^{\frac{n}{2}}} \exp\left(-\frac{1}{2\alpha^2} w^{\mathrm{T}}w\right)$$

n = Dimension of w

Learning as probabilistic model (Bayesian approach)

- **We want to find the posterior**

$$p(w|\mathbf{X}, y) \propto p(y|\mathbf{X}, w)p(w) = \frac{1}{(2\pi\sigma^2)^{m/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{m}(y_i - w^{\mathrm{T}}x_i)^2\right)\frac{1}{(2\pi\alpha^2)^{k/2}} \exp\left(-\frac{1}{2\alpha^2}w^{\mathrm{T}}w\right)$$

- **Take log:**

$$\begin{aligned}\log p(w|\mathbf{X}, y) &= -\frac{1}{2\sigma^2}\sum_{i=1}^{m}(y_i - w^{\mathrm{T}}x_i)^2 - \frac{1}{2\alpha^2}w^{\mathrm{T}}w + \mathrm{const} \\ &= -\frac{1}{2\sigma^2}\sum_{i=1}^{m} y_i^2 + \frac{1}{\sigma^2}\sum_{i=1}^{m} y_i x_i^{\mathrm{T}} w - \frac{1}{2\sigma^2}\sum_{i=1}^{m} w^{\mathrm{T}} x_i x_i^{\mathrm{T}} w - \frac{1}{2\alpha^2}w^{\mathrm{T}}w + \mathrm{const} \\ &= -\frac{1}{2\sigma^2}y^{\mathrm{T}}y + \frac{1}{\sigma^2}y^{\mathrm{T}}\mathbf{X}w - \frac{1}{2\sigma^2}w^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}w - \frac{1}{2\alpha^2}w^{\mathrm{T}}w + \mathrm{const} \\ &= -\frac{1}{2\sigma^2}y^{\mathrm{T}}y + \frac{1}{\sigma^2}y^{\mathrm{T}}\mathbf{X}w - \frac{1}{2}w^{\mathrm{T}}\left[\frac{1}{\sigma^2}\mathbf{X}^{\mathrm{T}}\mathbf{X} + \frac{1}{\alpha^2}\mathbf{I}\right]w + \mathrm{const}\end{aligned}$$

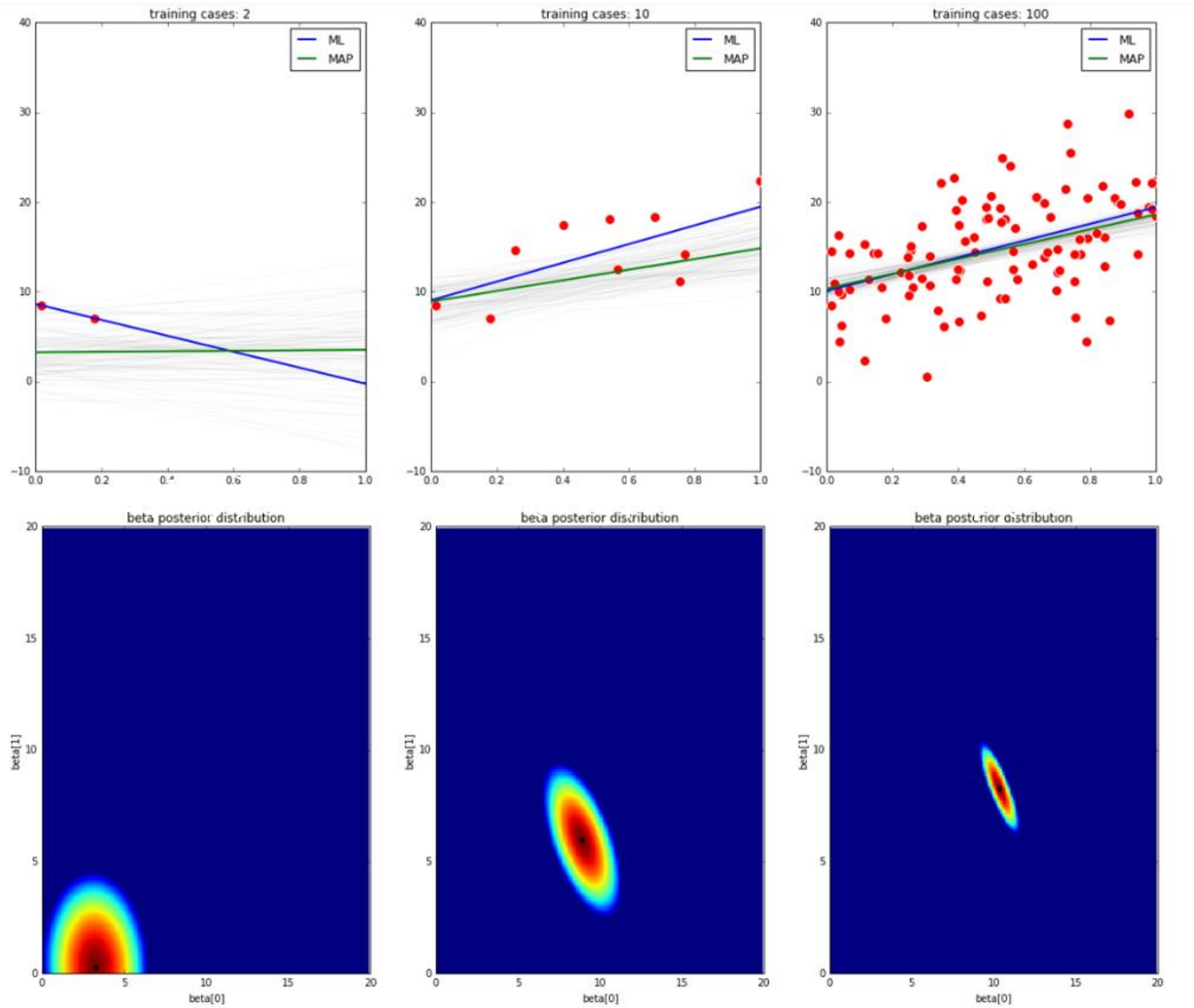
- **Posterior distribution is**

$$p(w|\mathbf{X}, y) = \mathrm{N}(w|\mu_w, \Sigma_w)$$

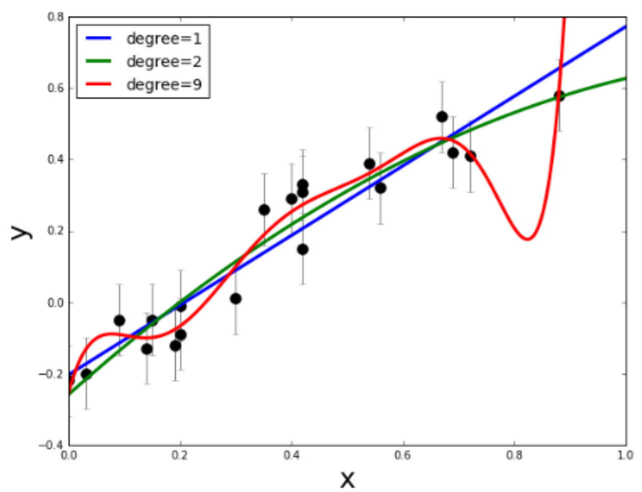
$$\mu_w = \Sigma_w\left(\frac{1}{\sigma^2}\mathbf{X}^{\mathrm{T}}y\right), \qquad \Sigma_w = \left[\frac{1}{\sigma^2}\mathbf{X}^{\mathrm{T}}\mathbf{X} + \frac{1}{\alpha^2}\mathbf{I}\right]^{-1}$$

Learning as probabilistic model (Bayesian approach)

- Draw sample $w \sim p(w|\mathbf{X}, y) = \mathrm{N}(w|\mu_w, \Sigma_w)$
- Draw sample $y = wx$



Regularized linear regression



What is a good regression function?

- The goal of regression is to come up with some good prediction function:

$$\hat{f}(x) = \widehat{w}^{\mathrm{T}} x$$

- So far, we have found $\widehat{w}$ by finding (Ordinary Least Square Estimation)

$$\widehat{w} = \underset{w}{\operatorname{argmin}} J(w) = \underset{w}{\operatorname{argmin}} \frac{1}{2} \|y - w\mathbf{X}\|_2^2$$

- To see $\hat{f}(x)$ is good candidate, we need to check
  - ✓ Is $\widehat{w}$ close to the true w
  - ✓ Will $\hat{f}(x)$ fit future observation well? **(Generalization)**

Regularized linear regression

- **Ordinary Least Square Estimation**
- OLS estimates the find the parameter that minimize the bias between the predicted and true values:

$$\hat{w} = \underset{w}{\operatorname{argmin}} \|y - w\mathbf{X}\|_2^2$$

- OLS estimates often have low bias but large variance
  → Poor generalization toward unseen test data set
- All features have a weight
  → Smaller subset with strong effects is more interpretable
- w_i's are unconstrained
  → They can explode and hence are susceptible to very high variance

We need some shrinkage (or regulation) to constraint $\hat{w}$

Regularized linear regression

Ridge regression

- Ridge regression introduces a regularization with the L-2 norm:

$$\hat{w} = \underset{w}{\operatorname{argmin}} \|y - w\mathbf{X}\|_2^2 + \lambda_2 \|w\|_2^2 \qquad \|w\|_2 = \sqrt{\Sigma_{i=1}^{k} w_i^2},$$

- Sacrifice a little of bias to reduce the variance of predicted values

→ More stable and generalize better

- Keep all the repressors in the model

→ Not easily interpretable model

Lasso (Least Absolute Shrinkage and Selection Operator)

- Lasso regression introduces a regularization with the L-1 norm:

$$\hat{w} = \underset{w}{\operatorname{argmin}} \|y - w\mathbf{X}\|_2^2 + \lambda_1 \|w\|_1 \qquad \|w\|_1 = \Sigma_{i=1}^{k} |w_i|$$

- Only a small subset of features with $\hat{w}_i \neq 0$ are selected

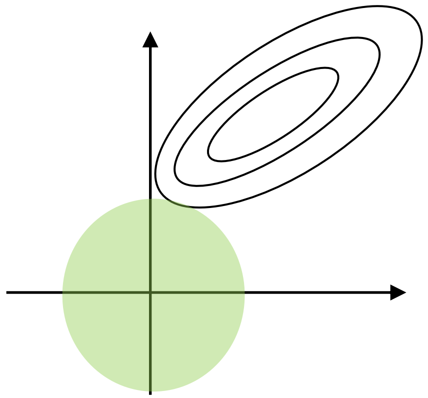
→ Increases the interpretability

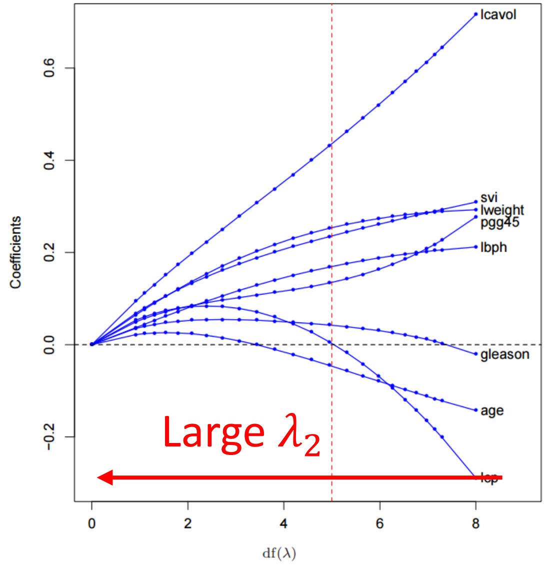
- More difficult to implement than Ridge Regression

Regularized linear regression

Ridge regression

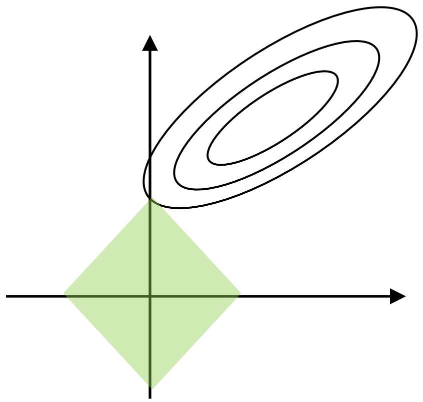
$$\hat{w} = \underset{w}{\mathrm{argmin}} \|y - w\mathbf{X}\|_2^2 + \lambda_2 \|w\|_2^2$$

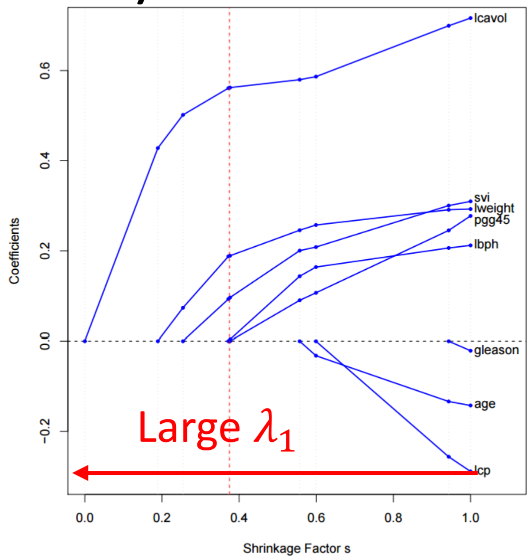




Lasso (Least Absolute Shrinkage and Selection Operator)

$$\hat{w} = \underset{w}{\mathrm{argmin}} \|y - w\mathbf{X}\|_2^2 + \lambda_1 \|w\|_1$$

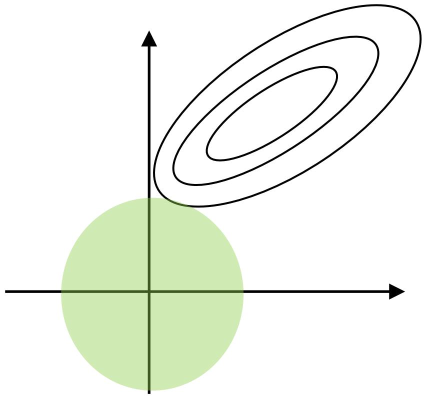




Bayesian view on regularized regression

Ridge regression

$$\hat{w} = \underset{w}{\operatorname{argmin}} \|y - w\mathbf{X}\|_2^2 + \lambda_2 \|w\|_2^2$$



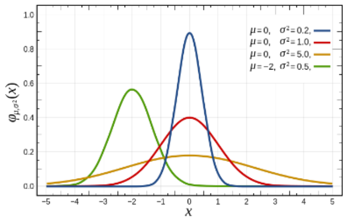
MAP estimation view

MAP estimation view

$$\hat{w} = \underset{w}{\operatorname{argmax}} \log p(w|\mathbf{X}, y) = p(y|\mathbf{X}, w)p(w)$$

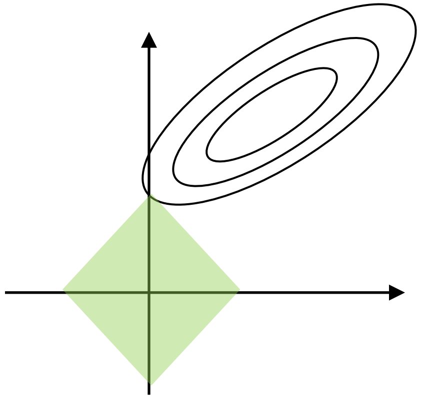
Gaussian prior

$$p(w) = \frac{1}{(2\pi\alpha^2)^{k/2}} \exp\left(-\frac{1}{2\alpha^2} w^{\mathrm{T}} w\right)$$



Lasso (Least Absolute Shrinkage and Selection Operator)

$$\hat{w} = \underset{w}{\operatorname{argmin}} \|y - w\mathbf{X}\|_2^2 + \lambda_1 \|w\|_1$$

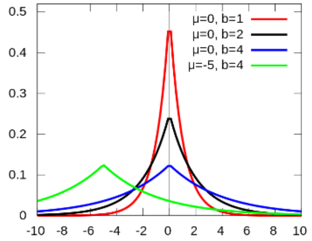


MAP estimation view

$$\hat{w} = \underset{w}{\operatorname{argmax}} \log p(w|\mathbf{X}, y) = p(y|\mathbf{X}, w)p(w)$$

Laplace prior

$$p(w) = \prod_{i=1}^{k} \frac{\lambda}{2\sqrt{\tau^2}} \exp\left(-\frac{\lambda |w_i|}{\sqrt{\tau^2}}\right)$$



Bayesian view on Ridge regression

m : number of data points
n : dimension of w

$$p(y|\mathbf{X}, w) = \prod_{i=1}^{m} \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{(y_i - w^{\mathrm{T}}x_i)^2}{2\sigma^2}\right)$$

$$p(w) = \mathrm{N}(w|\mathbf{0}, \tau^2\mathbf{I}) = \prod_{i=1}^{n} \frac{1}{(2\pi\tau^2)^{1/2}} \exp\left(-\frac{(w_i - 0)^2}{2\tau^2}\right) = \frac{1}{(2\pi\tau^2)^{n/2}} \exp\left(-\frac{1}{2\tau^2} w^{\mathrm{T}}w\right)$$

$$\begin{aligned} p(w|\mathbf{X}, y) &\propto p(y|\mathbf{X}, w)p(w) \\ &= \prod_{i=1}^{m} \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{(y_i - w^{\mathrm{T}}x_i)^2}{2\sigma^2}\right) \frac{1}{(2\pi\tau^2)^{k/2}} \exp\left(-\frac{1}{2\tau^2} w^{\mathrm{T}}w\right) \\ &= \left(\frac{1}{(2\pi\sigma^2)^{1/2}}\right)^m \frac{1}{(2\pi\tau^2)^{n/2}} \exp\left(-\sum_{i=1}^{m} \frac{(y_i - w^{\mathrm{T}}x_i)^2}{2\sigma^2} - \frac{1}{2\tau^2} w^{\mathrm{T}}w\right) \end{aligned}$$

$$\log p(w|\mathbf{X}, y) = m \log \frac{1}{(2\pi\sigma^2)^{1/2}} + \log \frac{1}{(2\pi\tau^2)^{n/2}} - \frac{1}{2\sigma^2}\left(\sum_{i=1}^{m} (y_i - w^{\mathrm{T}}x_i)^2 + \frac{\sigma^2}{\tau^2} w^{\mathrm{T}}w\right)$$

Maximum A Posteriori (MAP) estimation with Gaussian prior = **Ridge Regression**

$$(w^*) = \underset{w}{\operatorname{argmax}} \log p(w|\mathbf{X}, y) = \underset{w}{\operatorname{argmin}} \|y - w\mathbf{X}\|_2^2 + \lambda_2 \|w\|_2^2$$

Bayesian view on Lasso regression

$$p(y|\mathbf{X},w) = \prod_{i=1}^{m} \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{(y_i - w^{\mathrm{T}}x_i)^2}{2\sigma^2}\right)$$

m : number of data points
n : dimension of w

$$p(w) = \mathrm{Lap}(w|\lambda,\tau) = \prod_{i=1}^{n} \frac{\lambda}{2\sqrt{\tau^2}} \exp\left(-\frac{\lambda|w_i|}{\sqrt{\tau^2}}\right)$$

$$p(w|\mathbf{X},y) \propto p(y|\mathbf{X},w)p(w)$$

$$= \prod_{i=1}^{m} \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{(y_i - w^{\mathrm{T}}x_i)^2}{2\sigma^2}\right) \left(\frac{\lambda}{2\sqrt{\tau^2}}\right)^n \exp\left(-\frac{\lambda}{\sqrt{\tau^2}}\sum_{i=1}^{n}|w_i|\right)$$

$$= \left(\frac{1}{(2\pi\sigma^2)^{1/2}}\right)^m \left(\frac{\lambda}{2\sqrt{\tau^2}}\right)^n \exp\left(-\sum_{i=1}^{m}\frac{(y_i - w^{\mathrm{T}}x_i)^2}{2\sigma^2} - \frac{\lambda}{\sqrt{\tau^2}}\sum_{i=1}^{n}|w_i|\right)$$

$$\log p(w|\mathbf{X},y) = m\log\frac{1}{(2\pi\sigma^2)^{1/2}} + n\log\frac{\lambda}{2\sqrt{\tau^2}} - \frac{1}{2\sigma^2}\left(\sum_{i=1}^{m}(y_i - w^{\mathrm{T}}x_i)^2 + \frac{2\sigma^2\lambda}{\sqrt{\tau^2}}\sum_{i=1}^{n}|w_i|\right)$$

Maximum A Posteriori (MAP) estimation with Laplacian prior = **Lasso regression**

$$(w^*) = \underset{w}{\mathrm{argmax}}\log p(w|\mathbf{X},y) = \underset{w}{\mathrm{argmin}}\|y - w\mathbf{X}\|_2^2 + \lambda_1\|w\|_1$$

Questions?

?